Smart Solutions for Green Hydrogen

Using Waste to Clean Energy

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Abstract:

The Industrial Revolution initiated unprecedented technological advancements, yet it also contributed heavily to environmental degradation and climate change. In today's context, sustainable waste management has emerged as a critical area of focus to address these pressing concerns. Organic waste, when mismanaged, emits harmful gases; however, when properly utilized, it presents an opportunity to contribute to clean energy initiatives such as green hydrogen production.

This project introduces an intelligent and efficient system for estimating the weight of organic waste using deep learning and scientific modeling. The system follows a structured pipeline, beginning with the classification of waste type (organic or inorganic). For inorganic waste, a segmentation model is used to isolate objects from the background, followed by volume estimation based on pixel area and depth approximation. For organic waste, a classification model determines the object category and retrieves the corresponding mean density. In both cases, estimated volume and known density are used to calculate weight.

The system also includes a module for calculating the hydrogen ratio based on object type, as well as an economic module that estimates the market value of the material. This supports a transactional model in which users can sell their organic waste, based on its estimated hydrogen content and weight, to organizations engaged in green hydrogen production. All results are integrated into a user-friendly web platform designed for various stakeholders, including households, businesses, and environmental organizations, to visualize waste data, hydrogen potential, and associated economic metrics in real time.

In summary, this project offers a comprehensive, data-driven solution for sustainable organic waste management. By leveraging computer vision and web technologies, it facilitates accurate weight, value, and hydrogen content estimation, promoting informed decision-making and supporting global sustainability effort.

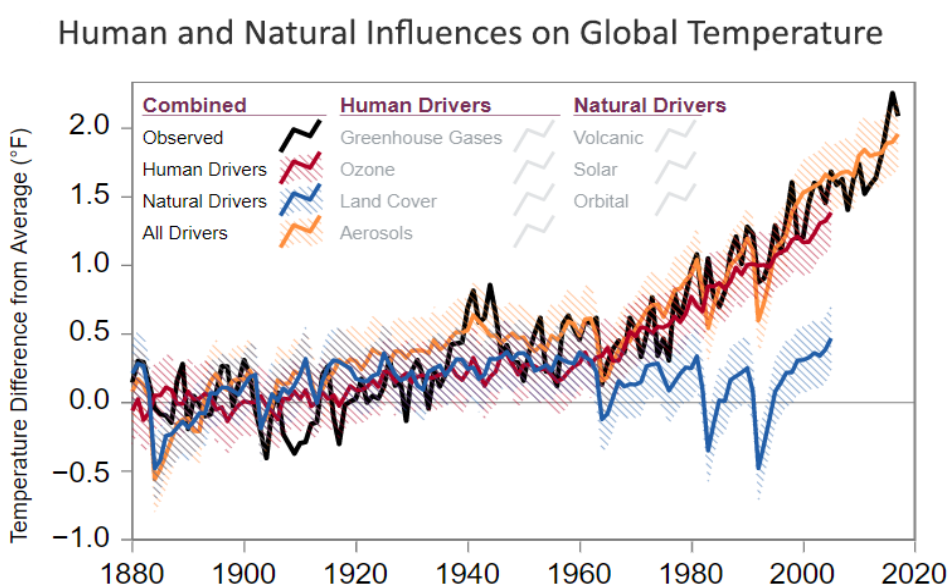
Chapter One

Introduction To the Project

1.1 Introduction:

Since the early nineteenth century, a prolonged change in temperature and weather patterns, referred to as climate change, has been occurring, potentially as a natural effect of fluctuations in the solar cycle [1]. Human activities have led to these changes, particularly the reliance on coal, oil, and gas as fossil fuels. The use of these fuels has triggered global warming due to the emission of greenhouse gases, like carbon dioxide and methane, resulting in increased earth temperatures, melting ice, severe weather events, alterations in ecosystems, ocean acidification, biodiversity loss, effects on agriculture and food security, along with problems related to water scarcity and quality. Energy is necessary for practically every area of modern life and is crucial to the smooth operation of our economy and society. It is estimated that the total amount of energy used worldwide will have increased significantly from 13 billion tons (BT) of oil equivalent in 2010 to 44 BT in 2050 [2]. The worldwide energy network will undergo a shift in the modern period, and "renewable energy (RE) production" is being considered as the modern and sustainable option. RE refers to any form of energy obtained from natural and renewable resources such as sunshine, wind, water, geothermal heat, and biomass [3].

Figure (1) Causes of Climate Change



Various processes and technological routes that employ both renewable and non-renewable feedstocks can produce hydrogen, such as natural gas, coal, nuclear, biomass, solar, wind, and waste [4]. The costs of renewable energy have fallen considerably in recent years and are expected to decline even more. Additionally, they are contrasting it with green hydrogen as a clean energy alternative, elucidating the characteristics of green hydrogen along with its production process and costs, as it is regarded as a future option for renewable energy sources [5].

In the face of increasing environmental challenges and the urgent need for sustainable energy solutions, green hydrogen has emerged as a promising alternative fuel source. Produced without emitting greenhouse gases, green hydrogen offers a cleaner pathway to address global energy demands. This project focuses on an innovative approach to determine the quantity of the extracted green hydrogen from organic waste, transforming discarded materials into a valuable resource.

This approach integrates circular economy principles by tackling two critical issues: managing organic waste and producing clean energy. By leveraging advanced technologies such as deep learning, weight modeling, and detection systems, this project aims to develop an efficient and practical solution for green hydrogen production. Deep learning techniques are employed to optimize critical aspects of the process, including waste classification, process efficiency, and predictive modeling, ensuring a robust and scalable system.

Our work not only advances renewable energy research but also demonstrates the potential of artificial intelligence in sustainable waste management. By combining cutting-edge technology with eco-friendly practices, this project sets the stage for innovative solutions that contribute to both clean energy production and environmental sustainability. This report details the project's objectives, methodologies, and anticipated outcomes, emphasizing the transformative role of deep learning and innovation in achieving sustainable solutions.

1.2 Problem Statement:

Hydrogen, as a clean and renewable energy source, has immense potential to drive sustainable development. Extracting hydrogen from organic waste not only provides an efficient way to produce energy but also addresses global challenges of waste management and environmental sustainability. However, accurately determining the quantity of hydrogen that can be produced from varying types of organic waste remains a significant challenge due to the complex and heterogeneous nature of waste materials.

Traditional methods for estimating hydrogen yield rely on biochemical or thermochemical analysis, which are often time-consuming, costly, and limited in their ability to generalize across different waste compositions. Variations in the type, moisture content, carbon-to-nitrogen ratio, and other properties of organic waste can significantly impact the efficiency and yield of hydrogen production processes. These challenges make it difficult for industries and researchers to optimize production systems and scale operations effectively.

Deep learning offers a promising solution to overcome these limitations by leveraging large datasets of organic waste characteristics and hydrogen yield results. Advanced neural networks can identify intricate relationships between waste properties and hydrogen output, enabling accurate prediction of hydrogen yield across diverse waste types.

Figure (2) Graphical abstract

This research focuses on developing a deep learning-based framework to accurately estimate the weight of organic waste and evaluate its potential for green hydrogen generation. The framework provides a reliable and scalable solution that bridges the gap between waste management and renewable energy through quantitative modeling and automation. By applying segmentation techniques, scientific weight estimation, and value assessment, the project contributes to global efforts in promoting sustainability and supporting the transition to clean energy.

1.3 Project Objective:

The goal of this project is to develop a sustainable and efficient system for green hydrogen production from organic waste by integrating advanced data science techniques and technologies. This project will not only address environmental challenges but also create a streamlined online platform to manage and automate the entire process of waste collection, analysis, and hydrogen production. The project follows a structured, step-by-step approach:

- **Collecting Images of Organic Waste**:

  The project will begin by collecting a diverse dataset of images representing various types of organic waste, including food waste, agricultural residues, and plant-based byproducts. These images will serve as the foundation for training deep learning models that will enable precise identification and classification of waste materials. A significant focus will be placed on obtaining high-quality, representative images from real-world sources.

- **Segmentation and Image Classification for Waste Identification**:

  Using deep learning algorithms, the project will develop segmentation models and image classification systems to crop and classify organic

waste within the collected images. Segmentation will separate relevant waste materials from the background, while classification models will categorize the waste types, such as food scraps or plant residues. This will ensure that only relevant data is processed for further analysis, improving the overall accuracy of the system.

- **Volume Estimation Modeling to Determine the Volume of Objects**:

Following classification, volume estimation is performed using a Mask R-CNN model fine-tuned on the UNIMIB2016 dataset, which is specialized for food-related object analysis. The model estimates the volume of each segmented object by analyzing mask areas and leveraging depth approximation techniques. This integrated process eliminates the need for 3D reconstruction while maintaining high accuracy in volume prediction, serving as a critical foundation for weight and hydrogen estimation.

- **Weight Calculation and Cost Estimation**:

Following the volume calculation, the project will estimate the weight of each waste object based on its material density and thickness. The weight will provide crucial information on the energy potential of the organic waste. In addition, the system will estimate the cost of purchasing the organic waste from users, factoring in transportation, collection, and processing costs. This will allow for a fair and transparent pricing structure, encouraging users to participate in the system and supply organic waste.

- **Development of an Online Platform for Real-Time Monitoring and Process Management**:

To integrate all these processes seamlessly, the project will create an interactive online platform. This platform will serve as the central hub for users, waste collectors, and system administrators. It will allow users to upload images of their organic waste, track real-time processing progress, and monitor key performance metrics like waste volume, hydrogen yield, and energy efficiency. The platform will also provide a dashboard for users

to receive cost estimates for their waste, offering transparency and motivating participation. The platform will automate key processes such as waste classification, Volume Estimation modeling, and cost estimation, making the entire workflow efficient and accessible online.

By combining data science, Deep learning model, and an interactive platform, this project aims to create a sustainable, efficient, and scalable solution for green hydrogen production from organic waste. The online platform will serve as a key enabler, connecting waste producers with the hydrogen production system, streamlining data collection and analysis, and facilitating real-time monitoring and decision-making. This approach will enhance both the environmental and economic sustainability of green hydrogen, contributing to the transition toward a low-carbon future.

1.4 Aim and Goal:

This project aims to explore and implement a sustainable approach for producing green hydrogen from organic waste, addressing both environmental and energy challenges. The focus is on developing an efficient and practical process that leverages modern technological tools, including segmentation models, volume estimation models, and weight models, to optimize the quantity of hydrogen extraction.

The primary goals of this project are:

- To **convert organic waste** into a clean energy source (green hydrogen) through a systematic and eco-friendly process.
- To **reduce environmental pollution** caused by organic waste, including methane emissions, and promote waste valorization.
- To **develop an efficient and scalable solution** that integrates advanced technologies for monitoring and improving the hydrogen production process.
- To **support the global shift toward renewable energy** by offering an alternative to fossil fuels.

- To **contribute to sustainable waste management** practices, providing a circular economy model that benefits both energy production and environmental preservation.
- To design and launch a **user-friendly website** where households, businesses, and organizations can easily sell their organic waste, fostering broader participation in waste management and creating economic incentives for sustainability.

The project aims to bridge the gap between **waste management challenges** and **clean energy demands** through these objectives, offering a viable solution for a greener future.

Figure (3) Waste Management System

1.5 Project Motivation:

The motivation behind this project stems from the need to address two critical global challenges: the environmental impact of organic waste and the urgent transition to sustainable energy. Improper disposal of organic waste contributes to significant environmental issues, such as greenhouse gas emissions and resource inefficiency. At the same time, **reliance on fossil fuels exacerbates climate change, threatening the planet's ecological balance.**

This project envisions transforming these challenges into opportunities by converting organic waste into green hydrogen, a clean and renewable energy source. This innovative approach aligns with the principles of the circular economy, offering dual benefits: **reducing waste-related environmental harm and supporting the shift toward renewable energy.**

Advanced technologies, such as **deep learning** and **weight estimation systems**, play a pivotal role in enhancing the efficiency, precision, and scalability of green hydrogen production. These tools enable real-time monitoring, process optimization, and breakthroughs in sustainable energy solutions, creating new possibilities for addressing global challenges.

Additionally, this project incorporates a user-friendly website designed to connect households, businesses, and organizations to the green hydrogen ecosystem. This platform will empower people to sell their organic waste with ease, fostering community participation and incentivizing waste management as a shared responsibility. By bridging waste generators and green energy producers, the platform promotes a collaborative model for sustainability.

The motivation driving this project extends beyond scientific and technological innovation. It is about making a tangible impact on communities, industries, and the environment. By demonstrating the potential of a sustainable and economically viable solution, this work aims to inspire a broader movement toward environmentally conscious practices, paving the way for a greener, more sustainable future.

We conducted a survey to assess public awareness regarding our project domain. The insights gained from this survey will help us enhance the relevance and effectiveness of our project.

1. How do you deal with your waste?

- **Get rid of waste**: Most common response, suggesting limited awareness or access to recycling.
- **Recycle it personally**: A minority practice, showing individual initiative.
- **Sell it**: Rarely chosen, indicating potential for growth in monetizing waste.

➢ **Insight**: There's significant room to encourage recycling and waste monetization.

After this simple question revealed striking insights 78.4% of respondents admitted to simply discarding waste without much thought, highlighting the pressing need for better waste management systems. Only 11.8% took steps to recycle, showing the limited reach of current recycling initiatives. Interestingly, 9.8% found creative, personal, or economical ways to handle waste, such as repurposing or selling it. These responses reflect a spectrum of behaviors and a clear opportunity to educate and empower individuals to make sustainable choices.

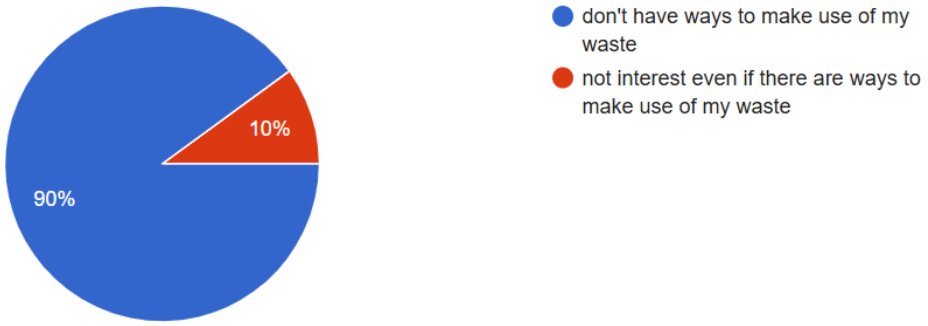
So, we make some questions for each case to treat with all categories of people.

In the first case (get rid of waste):

2. Why do you get rid of your waste?

- **Commonly cited:** Lack of awareness or methods to make use of waste.
- **Insight:** Highlighting the value of waste could change behaviors and we can benefit from this in developing sustainable development.

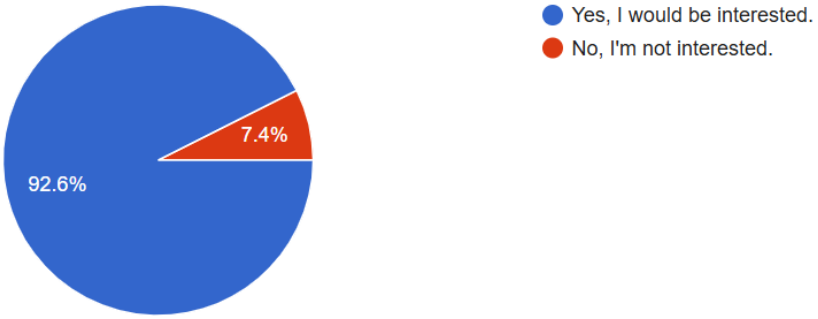
Why you get rid of your waste?



3. Would you be open to selling your waste in order to contribute to environmental sustainability?

- **Yes, I would be interested**: Common response, showing high openness.
- **No, I won't**: Minimal resistance.
- **Insight**: Positive attitudes toward sustainable waste management.

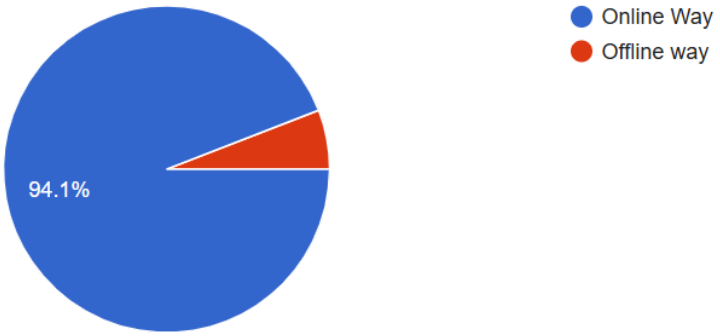
Would you be open to selling your waste in order to contribute to environmental sustainability? ***It could be a win-win situation*** 😉



4. If you have an online way to sell your waste, will you use it?

- ❖ **Yes, I will**: Most respondents are willing.
- ❖ **No**: Very few responses.
- ➢ **Insight**: Online platforms have high potential for adoption.

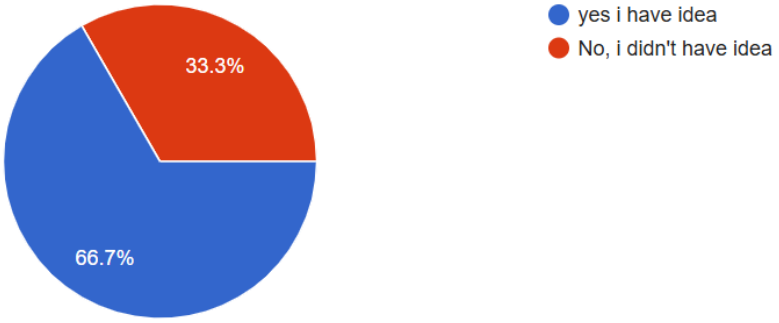
if there is a way to make use of your waste what you prefer to be?



5. Do you have an idea about organic and non-organic waste?

- ❖ **Yes, I have an idea**: Many respondents are aware of waste types, this will make it easier to us to spread the idea of project.
- ❖ **No, I didn't have an idea**: Awareness campaigns could target these individuals.
- ➢ **Insight**: While awareness exists, deeper education about waste segregation is needed.

do you have an idea about organic and non organic waste?

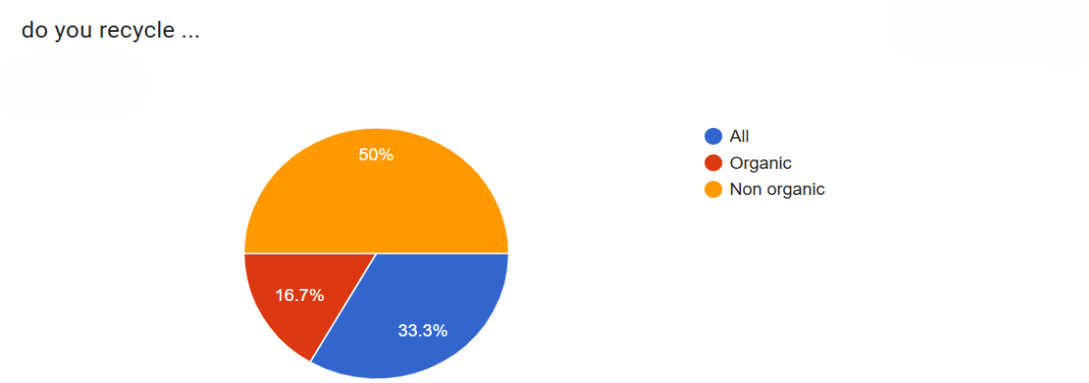


In the second case (Recycle it):

6. Do you recycle organic, non-organic, or all?

- **All**: Few respondents recycle comprehensively.
- **Organic:** Very few people are aware of recycle organic waste
- **Non-organic**: People only know how to recycle waste in old ways like oils, plastic, etc.

➢ **Insight**: Encouraging all-around recycling can improve sustainability.

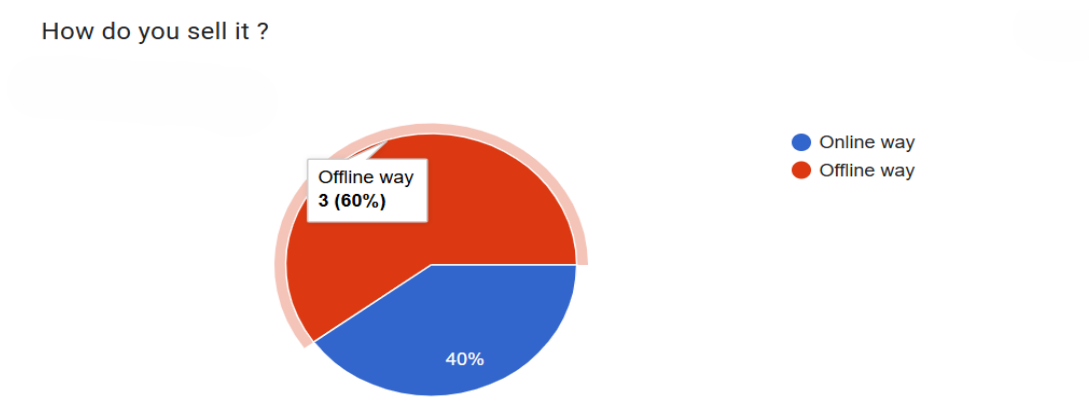


In the third case (sell it):

7. How do you sell it?

- **Offline Way**: Predominant choice.
- **Online Way**: Less frequent but promising for scalability.

➢ **Insight**: Promoting online platforms for waste selling could enhance accessibility and participation.



8. If you have an online way to sell your waste, will you use it?

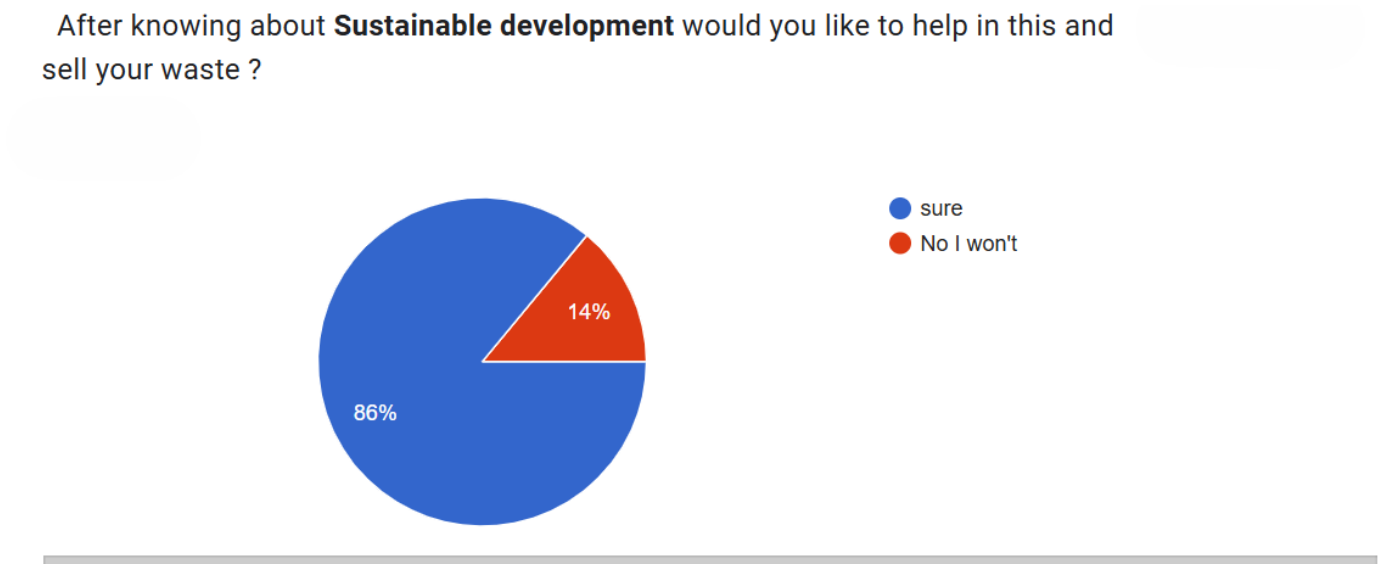
- **Yes, I will**: All respondents are willing.
- **No**: No responses prefer offline way.

➢ **Insight**: Online platforms have high potential for adoption.

9. After knowing about Sustainable Development, would you like to help in this and sell your waste?

- **Purpose**: Gauge interest in contributing to sustainability through waste management.
- **Response (Sure)**: Overwhelmingly positive response.

➢ **Insight**: Education on sustainability resonates with the audience.



1.6 Define User:

Types of Users

1. **Households**:

   - Everyday individuals and families who generate organic waste as part of their daily activities (e.g., food scraps, garden clippings).
   - Motivated by the desire to reduce waste, contribute to environmental sustainability, and potentially earn income by selling waste.

2. **Businesses**:

   - Small and medium enterprises (e.g., restaurants, and grocery stores) produce significant amounts of organic waste.
   - Driven by corporate social responsibility goals, cost-saving opportunities, and compliance with environmental regulations.

3. **Organizations**:

   - Larger institutions (e.g., municipalities, waste management companies) handling organic waste at scale.
   - Focused on optimizing waste collection, fostering community engagement, and advancing sustainability initiatives.

User Characteristics:

- **Environmentally Conscious**: Users are aware of the ecological impact of improper waste disposal and are willing to adopt practices that benefit the planet.
- **Technologically Open**: Ready to engage with innovative platforms, such as the project's user-friendly website, to manage waste effectively.
- **Financially Motivated**: Interested in monetizing organic waste, aligning sustainability with economic benefits.

- **Community-Oriented**: Keen on participating in broader efforts to promote sustainability and renewable energy within their communities.

User Goals:

- Simplify the process of waste disposal.
- Access reliable platforms to sell organic waste.
- Contribute to environmental sustainability efforts.
- Support renewable energy initiatives like green hydrogen production.

1.7 Challenges:

1. Data Collection and Preprocessing

- **Challenge**: Collecting a large and diverse set of images related to organic waste and green hydrogen production can be difficult, especially if high-quality labeled data is scarce. Cleaning and preparing this data for training models also requires time and resources to ensure it is accurate and representative.
- **Potential Impact**: Inaccurate or insufficient data could lead to poorly performing models and hinder the effectiveness of downstream tasks like segmentation and detection.

2. Modeling and Segmentation Accuracy

- **Challenge**: Developing accurate segmentation models that can precisely identify and separate relevant elements in images (e.g., different types of organic waste or energy-related components) is a complex task. It requires the models to be both highly accurate and robust to variations in lighting, image quality, and object positioning.
- **Potential Impact**: Inaccurate segmentation could lead to poor extraction of relevant features from the images, affecting the efficiency of green hydrogen production and waste management processes.

3. Classification Model

- **Challenge**: After successful segmentation of waste items, classifying each segmented object accurately is essential to assign the correct density and hydrogen ratio values. However, visual similarities between different types of organic waste and limited class diversity in the dataset can make classification challenging.
- **Potential Impact**: Misclassification can propagate errors in downstream computations, such as weight and hydrogen content estimation, leading to inaccuracies in resource valuation and user incentives.

4. Volume and Weight Estimation

- **Challenge**: The system uses a Mask R-CNN model (fine-tuned on the UNIMIB2016 dataset) to extract object masks and approximate object volume using pixel area and assumed depth. This method, while efficient, is sensitive to the quality of segmentation and relies on simplified geometric assumptions, which can limit its precision.
- **Potential Impact**: Errors in 3D modeling or weight calculation could affect resource allocation and planning for waste management, hindering the overall optimization of the green hydrogen production process.

5. Real-Time Processing and Scalability

- **Challenge**: Implementing real-time processing capabilities for image data on a scalable platform presents several challenges. This includes efficiently handling large volumes of data, processing images in real-time, and ensuring the models remain accurate and fast under heavy load.
- **Potential Impact**: Errors in volume estimation will directly impact weight calculation and all dependent modules such as hydrogen estimation and pricing, thereby affecting the overall performance and trustworthiness of the system.

6. Platform Development and Integration

- **Challenge**: Building and maintaining a user-friendly platform that integrates the complex models, data workflows, and user interactions is

a significant challenge. Ensuring smooth interaction between the frontend (user-facing side) and the backend (data models, processing) is crucial for a seamless experience.

- **Potential Impact**: A poorly designed platform could discourage users from participating, limiting the effectiveness of the waste-to-hydrogen ecosystem.

7. Model Generalization and Robustness

- **Challenge**: Ensuring that the models work effectively across a wide range of environments and scenarios, especially as new types of organic waste or unexpected situations arise, is key. The models need to generalize well to different lighting, quality, and types of data.
- **Potential Impact**: Models that fail to generalize could lead to low performance in real-world applications, limiting the project's ability to scale and contribute to sustainable energy solutions.

8. Data Privacy and Security

- **Challenge**: Since the platform involves user-generated content (e.g., images of organic waste), ensuring data privacy and security is critical. You'll need to comply with regulations and ensure that sensitive information is protected.
- **Potential Impact**: Data breaches or privacy concerns could harm the platform's credibility and reduce user trust, leading to lower engagement and possible legal issues.

Chapter Two

Review

[6] This paper combines YOLO for object detection with the Sliding Innovation Filter for robust tracking, improving accuracy and stability in dynamic environments. It is focused on addressing noise and uncertainties in detection and tracking tasks.

[7] The paper introduces YOLOv3, a real-time object detection model that improves speed and accuracy, supporting multi-scale predictions and the detection of smaller objects effectively.

[8] This research presents LC-YOLO, a compact model optimized for real-time object detection with limited computational resources, making it suitable for embedded systems and mobile applications.

[9] YOLO is introduced as a single convolutional neural network that directly predicts bounding boxes and class probabilities from input images, offering a fast and accurate real-time object detection framework.

[10] This paper evaluates the performance of YOLO in real-world scenarios using OpenCV, focusing on speed and accuracy improvements for detecting and tracking moving objects in video feeds.

[11] YOLO-World extends the YOLO model for open-vocabulary detection by integrating vision-language pretraining, enabling the identification of novel objects not included in the training data.

[12] This study evaluates YOLO's performance in large-scale object detection, analyzing its robustness under challenging ambient lighting conditions in images captured from network cameras.

[13] The paper reviews the applications of YOLO in agriculture, covering tasks such as crop monitoring, pest detection, and automation, while providing insights into trends and challenges in this domain.

[14] This paper introduces U-Net, one of the most popular architectures for image segmentation, and demonstrates its application in biomedical image segmentation. U-Net’s architecture can be adapted to classify and segment waste materials in the hydrogen extraction process.

[15] This paper presents Mask R-CNN, an extension of Faster R-CNN that adds a branch for predicting segmentation masks. It can be applied to 2D waste material segmentation.

[16] A paper that presents the 3D U-Net architecture for volumetric image segmentation. The model is effective in processing 3D data, making it suitable for 3D segmentation tasks in hydrogen extraction.

[17] This paper introduces PointNet, which directly processes point cloud data. It’s highly applicable for 3D waste classification and segmentation in hydrogen extraction processes.

[18] This paper introduces DeepLab, a semantic segmentation model that uses atrous convolutions. It's highly effective for semantic segmentation tasks such as waste material identification.

[19] Though focused on audio, this paper touches on the efficiency of convolutional neural networks (CNNs) in various domains. The insights can be adapted for image segmentation tasks like material classification in waste for hydrogen extraction.

[20] This paper discusses efficient methods for 3D deep learning applied to point cloud data, which is directly applicable to 3D segmentation of waste material for hydrogen extraction.

[21] This paper focuses on using CNNs for classifying waste materials. It discusses the potential of AI to automate waste management tasks, which can be directly applied to hydrogen extraction workflows.

[22] This study demonstrates the use of CNNs for the semantic segmentation of agricultural waste. It provides a useful framework for applying segmentation in industrial waste for the hydrogen extraction project.

[23] SegNet is an encoder-decoder architecture designed specifically for image segmentation. The approach could be useful for waste sorting and material identification in the hydrogen extraction process.

[24] This paper presents advanced self-supervised learning techniques to enhance monocular depth estimation accuracy. By introducing innovative loss functions and training strategies, the authors demonstrate significant improvements over previous self-supervised methods. The approach reduces reliance on labeled data while achieving competitive results, showcasing the potential of self-supervised learning for scalable depth estimation solutions.

[25] This paper pioneers the use of a multi-scale deep network for predicting depth maps from single images, a significant advancement in computer vision tasks like 3D scene reconstruction and autonomous driving. The authors propose a novel architecture that integrates global coarse predictions with refined local predictions, resulting in more accurate and detailed depth maps. The multi-scale approach leverages contextual information effectively, showcasing its superiority over traditional methods and setting a foundation for subsequent works in single-image depth estimation.

[26] The paper introduces a domain adaptation approach for real-time monocular depth estimation using synthetic datasets. By employing image style transfer, the authors successfully bridge the domain gap between synthetic and real-world data, enabling accurate depth predictions in real-time scenarios. The approach highlights the potential of leveraging synthetic data for training deep networks while addressing the challenges of domain adaptation effectively.

[27] This paper addresses the challenge of depth estimation from unconstrained, real-world images. Unlike previous methods relying on curated datasets, the authors propose a robust approach capable of handling diverse scenes, lighting conditions, and occlusions. By introducing a novel loss function and training strategy tailored for "in-the-

wild" images, the work makes substantial progress in bridging the gap between lab-based depth prediction methods and real-world applications.

[28] This paper explores unsupervised depth prediction from monocular video sequences, eliminating the dependency on sensor-based ground truth data. By leveraging scene structure and motion, the proposed framework introduces a novel self-supervised learning paradigm that achieves impressive depth prediction accuracy. The methodology demonstrates the potential of leveraging video data for robust and scalable depth estimation, opening new avenues for research in unsupervised learning.

[29] The paper introduces Exponential Linear Units (ELUs) as a novel activation function designed to accelerate deep network learning while maintaining high accuracy. Traditional activation functions, such as ReLU, often suffer from issues like gradient vanishing and dead neurons during training. ELUs address these limitations by combining the advantages of faster convergence with the ability to achieve better generalization. Through extensive experiments, the authors demonstrate that ELUs outperform other activation functions across various datasets and architectures, particularly in training deep networks efficiently and effectively.

[30] Building upon their earlier work, the authors present a unified framework capable of predicting depth, surface normals, and semantic labels simultaneously. This multi-task learning approach employs a common multi-scale convolutional architecture to optimize computational resources and improve prediction quality across tasks. The integration of these complementary tasks into a single network highlights the paper's innovative methodology, demonstrating significant improvements in accuracy and efficiency for depth estimation and related tasks.

[31] This work focuses on learning rigid body motion using SE3-Nets, a novel neural network architecture designed to model rigid

transformations. By directly predicting SE(3) parameters, the network captures the underlying geometric properties of rigid body motion effectively. The paper's contributions lie in its ability to learn motion dynamics from raw sensory inputs, offering valuable insights for robotics and autonomous systems applications.

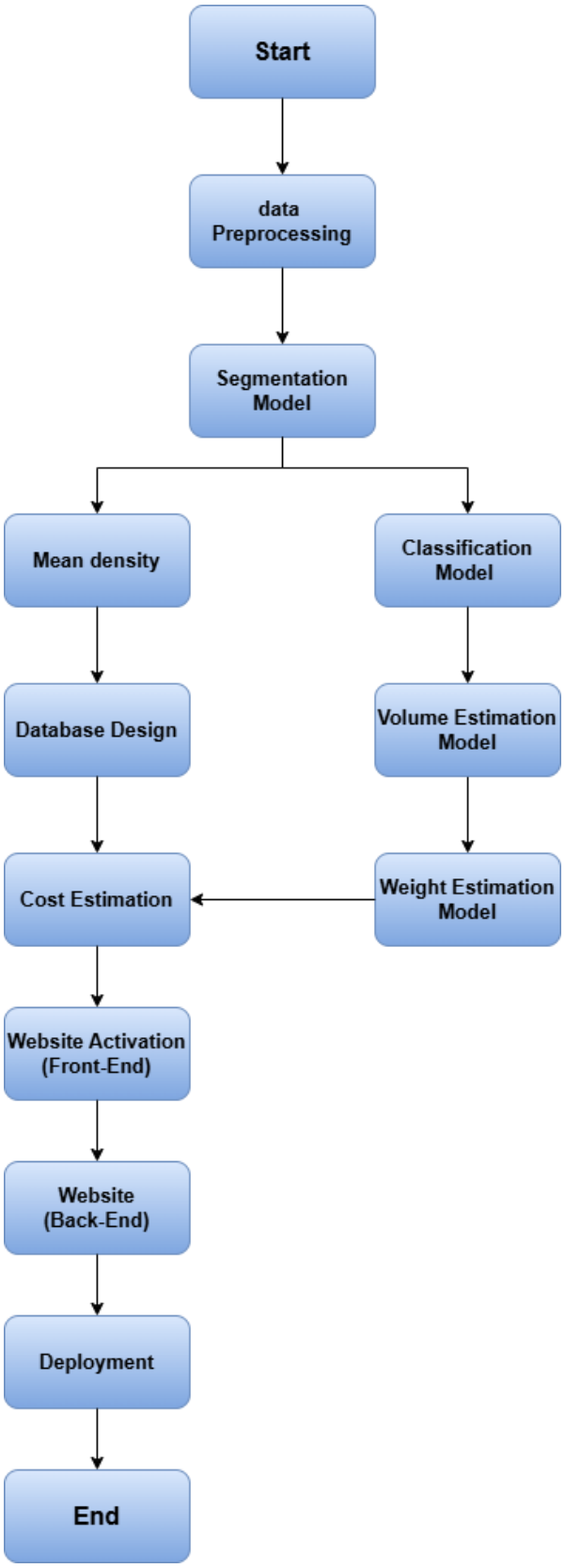
[32] The authors propose a multi-scale modeling framework to improve the accuracy and robustness of monocular depth estimation. By combining fine-grained local features with global contextual information, the framework effectively captures complex scene structures. The experimental results demonstrate the superiority of this approach, particularly in challenging scenarios involving occlusions and diverse scene compositions.

[33] This paper provides a comprehensive overview of recent advancements in deep learning models for image-based depth prediction. The authors analyze various architectures, loss functions, and datasets, highlighting the strengths and limitations of each approach. Furthermore, the paper discusses practical applications of depth prediction, offering valuable insights for researchers and practitioners in fields such as augmented reality, robotics, and autonomous vehicles

Chapter Three

Methodology

Figure (4) System Process Flowchart



3.1 System Request:

System Request:

The project aims to provide a scalable, automated solution for hydrogen extraction from waste. This contributes to clean energy innovation by addressing waste management and energy sustainability simultaneously.

3.2 Dataset Description:

The TACO (Trash Annotations in Context) dataset is an open-source collection designed to advance research in litter detection and segmentation. It comprises images of waste captured in diverse real-world settings, including woods, roads, and beaches. Each image is meticulously annotated to facilitate the development and evaluation of object detection algorithms.

Key Features:

- **Diverse Environments**: The dataset includes images from various settings, such as woods, roads, and beaches, providing a comprehensive representation of litter in different contexts.
- **Detailed Annotations**: Images are manually labeled and segmented according to a hierarchical taxonomy, enabling precise training and evaluation of object detection algorithms.
- **Growing Collection**: TACO is continually expanding through crowdsourcing, with new images and annotations added regularly to enhance its utility for research and development.

Applications:

TACO supports various research areas, including:

- **Litter Detection**: Identifying and classifying different types of waste in images to aid in environmental monitoring and cleanup efforts.

- **Object Detection Algorithm Development**: Providing a benchmark for developing and testing algorithms aimed at detecting and segmenting litter in diverse contexts.

By offering a rich and varied dataset, TACO enables the development of models that can generalize across a wide range of real-world litter detection scenarios.

3.3 Process Structure:

The pipeline includes the following stages:

1. **Data Preprocessing:**
    - **Exploration Phase**: This involves initial inspection of the dataset to identify key patterns, outliers, and relationships. Using visualization tools, the team analyzed the diversity of waste types in the TACO dataset, ensuring the data represented various environmental contexts like urban areas, beaches, and forests. This step also included identifying potential biases in the dataset.
    - **Augmentation Phase**: The augmentation process utilized Roboflow to apply transformations such as rotation, scaling, brightness, and contrast adjustments. Roboflow's advanced augmentation tools streamlined the creation of diverse representations of the same objects. This enriched the dataset and significantly enhanced the model's ability to generalize across unseen scenarios by simulating real-world variations effectively.
2. **Segmentation Model**:
    - **Purpose**: The segmentation model’s primary role is to detect and segment waste objects within images accurately.

- **Model Selection**: The project employed the YOLOv11x model, fine-tuned specifically for waste detection tasks using the TACO dataset. Initially, the model achieved a low accuracy of 30%. However, through the integration of Bayesian Search for hyperparameter tuning and the application of advanced augmentation techniques, the accuracy was improved significantly, reaching 57.3%. The YOLOv11x model was further enhanced by leveraging an SGD optimizer, ensuring improved gradient updates and performance stability during training.The training process began with baseline parameters, and Bayesian Search was implemented to identify optimal configurations for learning rates, batch sizes, and anchor box settings. Augmentation techniques introduced variations in object orientation, lighting, and size to reduce overfitting and improve robustness.A comprehensive evaluation highlighted areas needing refinement, such as missed detections in cluttered environments, which were iteratively addressed.

- **Tools Used**: The team utilized Roboflow for preprocessing annotations, organizing datasets, and managing image manipulations. PyTorch served as the framework for model training and deployment. Detailed performance evaluation focused on precision, recall, and F1 score, driving iterative improvements in segmentation accuracy.

3. **Convex Hull Model**:

- **Purpose**: This model addresses incomplete or overlapping segments produced by the segmentation phase. By reconstructing shapes, it ensures better representation of individual objects for subsequent analysis.

- **Approach:** A convex hull algorithm was applied to approximate the geometric boundaries of segmented objects, smoothing irregular shapes and filling gaps. The algorithm focused on identifying the smallest convex shape encompassing all points of a detected object to account for irregularities effectively. Validation involved overlaying reconstructed shapes onto original images to ensure consistency and testing against manually corrected samples to confirm reliability. Additional adjustments were incorporated to handle complex geometries, such as concave edges, ensuring minimal loss of detail during reconstruction.
- **Tools Used**: OpenCV was used for implementing and testing the convex hull computations, supported by NumPy for data handling and visualization. Matplotlib was used for overlay visualization and accuracy assessments, providing insights into performance during development.

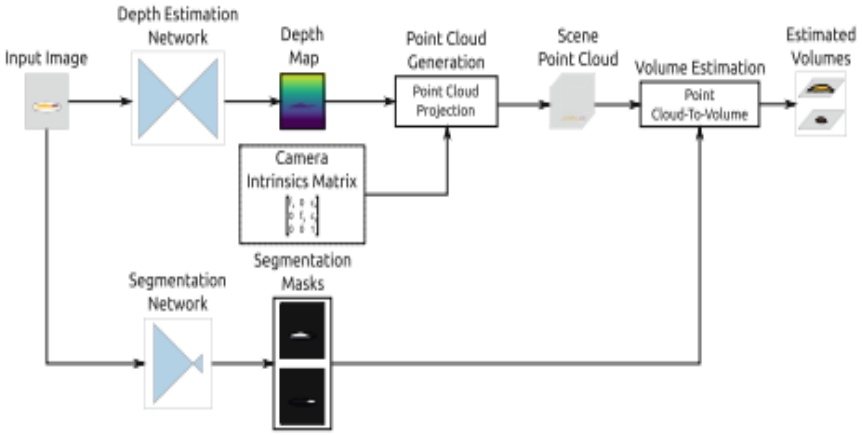
4. **Object Classification**:
    - **Purpose**: To classify segmented objects into predefined material categories (e.g., food types like beef, pudding, pie) in order to enable accurate downstream calculations such as weight, hydrogen ratio, and price estimation.
    - **Approach**: This stage involved applying a classification model to each segmented mask generated by the segmentation model (Mask R-CNN). The classifier leverages deep convolutional neural networks to extract and analyze visual features specific to each waste type. A fine-tuned CNN was trained on the cropped segments to associate each object with its respective material class, allowing for automatic mapping to average density and hydrogen ratio values. Attention mechanisms were incorporated to emphasize the most relevant spatial features and reduce noise from irrelevant image regions.

- **Enhancements**: To enhance model performance with limited labeled data, a semi-supervised learning strategy was adopted. Pseudo-labeling was used to expand the training set by assigning predicted labels to high-confidence unlabeled data, which were then reintroduced into the training pipeline, thereby improving generalization.
- **Tools Used:** The classification models were implemented in **PyTorch**, with transfer learning applied using pretrained CNN backbones for faster convergence and better feature representation. **Roboflow** was utilized for preprocessing, automated cropping of segmented objects, and augmentation. Evaluation was conducted using **scikit-learn** to compute precision, recall, and F1-score, ensuring reliable assessment of classification quality.

5. **Volume Estimation:**

- The proposed food volume estimation system can be separated into three distinct parts:
  - the depth estimation network
  - the segmentation network and
  - the point cloud-to-volume algorithm

Figure (5) Multi-channel depth segmentation



Depth Estimation Network:

The depth network architecture used in the proposed system is the one presented in the work of Godard, in which they demonstrate training a depth estimation network using only monocular video sequences. In each training step, three consecutive frames I_{t-1}, l_t,$I_{t+1}$of the video is used. **The depth prediction network** infers a depth map D_t of the input frame l_t, while a **pose estimation network** generates the camera pose transformations $T_{t-1\to t}$, $T_{t+1\to t}$between the current and its adjacent frames. The predicted depth map, pose transformations and known camera intrinsic matrix K are used to synthesize the center frame by sampling from the previous and next as

$I_{t-1\to t}$ = $I_{t-1} \ll proj(D_t, T_{t\to t-1}, K) \gg$

$I_{t+1\to t}$ = $I_{t+1} \ll proj(D_t, T_{t\to t+1}, K) \gg$

where $proj$ is the coordinate projection and the sampling operator. The final training loss is the sum of a photometric loss $L_p$between the synthesized and original images and a depth smoothness error L_s, given as

Figure (6) Monocular depth estimation

$L = L_p + \lambda L_s$

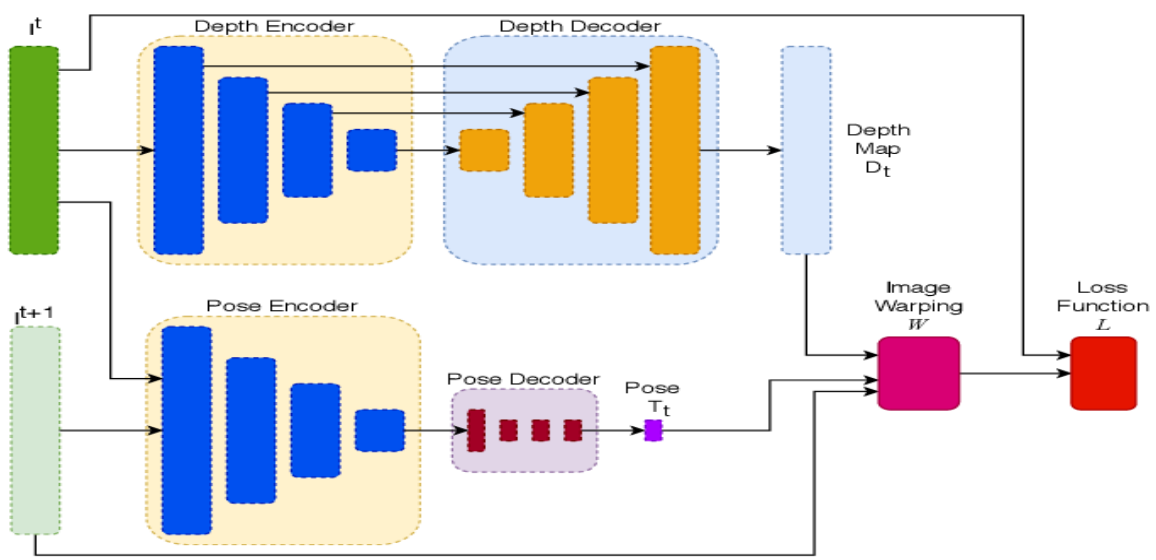


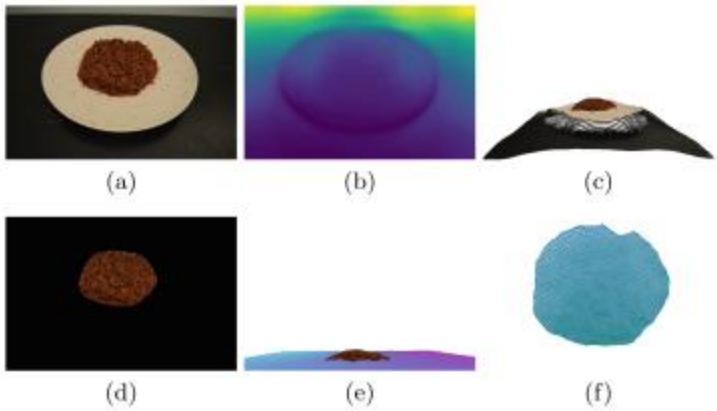
Fig. 2. Monocular depth estimation pipeline [2]. Modules enabled for instance adaptation during inference marked with yellow background

This approach fails to generate a meaningful training signal when the true pose transformations ${T^{gt}}_{t\to t-1}$, ${T^{gt}}_{t\to t-1}$ are zero, since in this case, the predicted depth values do not affect the image synthesis whatsoever. This limits the videos we can train the network with to those having sufficient motion between frames, which is not as common among food videos, since they are mostly captured from stationary cameras. We train the network on the EPIC-KITCHENS dataset, which includes more than fifty hours of egocentric food-handling videos. The selected cooking and meal preparation sequences were taken from six different kitchens, with the consideration of incorporating variations in the environment, actions and lighting conditions. This dataset, despite being sub-optimal for the task, with it focusing around the miscellaneous actions of preparing food in a kitchen, was chosen in lieu of other food-specific video datasets, due to being the only one to satisfy the camera motion limitation. For instance, in the Pittsburgh fast-food image dataset, the authors employ a stationary camera and a rotating platter to capture their videos, which would not be suitable for training with this architecture. To further improve the depth predictions, we fine-tune the network weights on a dataset of food-specific videos we collected, which we will refer to as food videos. In total, 38 videos were captured on daily occasions, using commercial smartphone cameras and portray a number of different food types and environments. Training with this dataset further backs the argument on the ease of training depth estimation networks in a self-supervised manner, considering that gathering an equivalent dataset of food-image depth data would require substantially more resources and time.

Food Segmentation Network

To be able to locate the food objects present in a given image, we train an instance segmentation network on a dataset of labeled food images. The model utilized for this purpose is the Mask R-CNN

Figure (7) Segmentation network in food image



extends the object detection Fast Region-based CNN architecture, by additionally predicting individual segmentation masks for each object found. With this approach, we are able to discern between the different foods present in the input, including multiple instances of the same food type and produce a segmentation mask for each, to estimate their volume separately. We initialize the network using weights pre-trained on the COCO dataset and fine-tune for food object segmentation on the UNIMIB2016 dataset . The later, is composed of 1,027 meal tray images, labeled with bounding boxes, food types and segmentation masks for the food objects depicted. Since we are not interested in predicting the specific food types, we ignore the food-type label

System stages of estimating the volume of a food object. (a) Input image. (b) Depth prediction. (c) Point cloud representation of the input. (d) Generated segmentation mask. (e) Base plane estimation. (f) Triangulation

Volume Estimation

Having inferred the depth map D of the input image and given the camera intrinsic matrix K, we project each pixel (x, y) to a corresponding point in 3D space, using its homogeneous coordinates and the inverse projection model

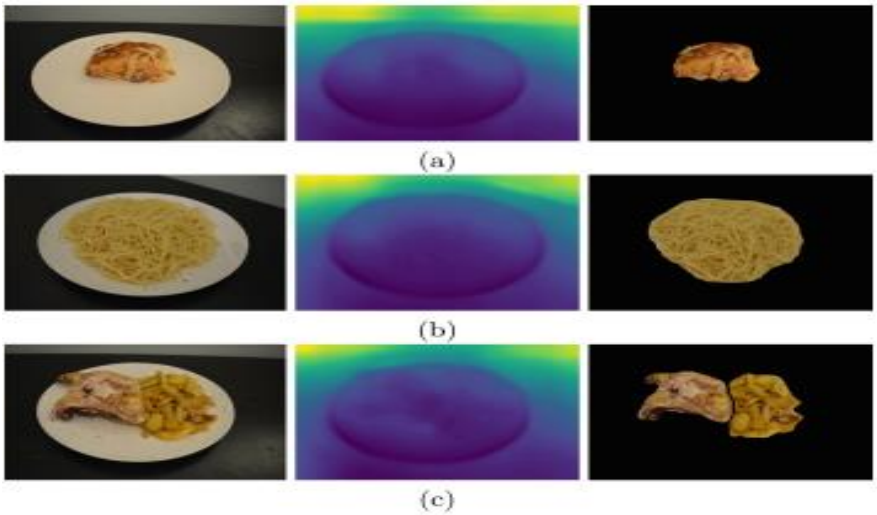
P_{xy}=$K^{-1}[xy1]^T D_{xy}$,

to form a point cloud representation P. In turn, the segmentation masks generated by the instance segmentation network, are applied onto the image to distinguish between the different foods depicted and split P into subsets of food object points.

For each of these sets of points, we first remove any outliers using a statistical outlier removal (SOR) filter and then determine the base plane on which the food is placed upon by applying principal component analysis. The eigenvector of the least important component is set as the plane normal vector, while the plane is also adjusted to be at the bottom of the object. The final step for approximating the volume contained between the base and food is projecting the food points onto the plane and partitioning the covered area into triangles by computing an α-complex from the Delaunay triangulation. The total volume is made up of the triangular prisms, defined by each triangle and the average distance of its vertices from their corresponding food points. The stages of estimating the volume of a food object in an image.

This approach only works in cases where the food is placed on top of a planar plate. We cannot estimate the volume of foods that are inside containers, such as a bowl, in which case the volume is not defined by the food object, but instead by the container itself.

Figure (8) Test set used for evaluation



Samples from the test set used for evaluation, along with the depth and segmentation mask predictions.

- Sample from the Souffle food type.
- Sample from the Spaghetti food type.
- Sample from the Potatoes /Chicken food type

Analysis:

- **Purpose**: Analyze physical and chemical characteristics of detected objects, including volume estimation and mass prediction. This data informs hydrogen yield simulations.
- **Approach**: Advanced simulations were conducted to assess chemical properties and estimate hydrogen output. Various machine learning techniques, including regression models and clustering, were employed to refine predictions of hydrogen yield potential based on object properties. Feature engineering techniques identified correlations between object properties and hydrogen yield potential, leveraging extensive exploratory data analysis to enhance accuracy.
- **Tools Used**: Libraries like Pandas and NumPy enabled efficient data manipulation, while Seaborn facilitated detailed visual analysis. Python-based frameworks such as Scikit-learn were also employed for building predictive models, further enhancing hydrogen output estimations.

Statistical Model:

- **Purpose**: To calculate precise ranges for key parameters like volume, mass, and hydrogen percentage, enhancing prediction accuracy.
- **Approach**: Statistical inference methods established confidence intervals and identified variability across samples. Robust data stratification techniques were used to ensure insights were applicable across varying waste compositions. Metrics like mass-to-hydrogen conversion efficiency were evaluated for different waste categories, using advanced models to estimate outlier impact and refine conversions. Uncertainty quantification was prioritized to highlight confidence levels in hydrogen yield predictions, aligning simulations closer to real-world data constraints.
- **Tools Used**: SciPy provided robust statistical computation capabilities, supported by NumPy for data structuring and analysis. Supplementary visualizations created via Matplotlib provided insights into parameter distributions and comparative metrics.

Testing and Deployment:

- **Unit and Integration Tests**: Both unit tests and integration tests were conducted to ensure individual components functioned correctly and worked seamlessly together. This ensured robustness and reliability across all system components.
- **Web Application Deployment**:
    - **Frontend**: React.js was used to create an interactive and responsive web interface. The application's first page allows users to upload or drag images, take a photo, and explore an "About" section for an overview of the project. The second page provides real-time processing results, displaying a summary of model outputs and detailed information in a structured table.

- **Backend**: FastAPI managed backend functionalities, handling requests, processing data, and interfacing with models. It ensured low-latency responses for seamless user experiences.
- **Hosting**: Railway (formerly Heroku) was used for hosting, selected for its simplicity, scalability, and support for continuous integration and deployment. Its robust infrastructure enabled the app to handle global accessibility efficiently.
- **Modularity**: Both the React.js and FastAPI components were designed for modularity, allowing updates without significant system overhauls.

Spiral Model

The Spiral Model has been chosen as the methodology for this project. This iterative approach allows for continuous refinement and risk assessment at each stage of development, which is crucial for a project involving advanced technologies and experimental methodologies.

Reasons for Choosing the Spiral Model:

1. **Iterative Development**: The project's complexity, involving segmentation, convex hull modeling, and statistical analysis, requires iterative cycles to ensure accurate results.
2. **Risk Management**: Each stage of the process (e.g., preprocessing, segmentation, deployment) is prone to unique risks, such as overfitting or data inconsistencies. The Spiral Model's emphasis on risk analysis mitigates these issues early.
3. **Flexibility**: As the project incorporates diverse tools and techniques, the Spiral Model accommodates changes and enhancements seamlessly during development.
4. **Progressive Refinement**: This methodology supports gradual improvement through feedback and validation, ensuring that the final system meets both functional and performance requirements.

5. **Stakeholder Involvement**: Regular reviews at each spiral cycle allow for continuous alignment with project objectives and stakeholder expectations.

By employing the Spiral Model, this project can adapt dynamically to challenges, optimize resources, and deliver a robust solution for green hydrogen extraction using waste.

3.4 Timeline Schedule:

Week	Task	Status
1-3	Data Preprocessing	Completed
4-6	Segmentation Model	Completed
7	UI/UX Design	Completed
8-10	Convex Hull Model Development and Testing	In Progress
11-12	Classification Model Training	Planned
13-16	Volume and Weight Estimation	Planned
17-18	Analysis	Planned
19	Statistical Model Development	Planned
20-23	Testing, Deployment, and Final Optimization	Planned

Chapter 4

Implementation

4.1 Data Preprocessing

Exploration Phase

The data preprocessing phase began with a thorough exploratory analysis of the TACO (Trash Annotations in Context) dataset. The primary objective was to understand the underlying structure and quality of the data before feeding it into the modeling pipeline.

During this stage:

- Visual inspection and summary statistics were used to identify key patterns, such as the distribution of waste categories, object sizes, and contextual diversity.

- Environmental variety within the dataset was confirmed, including samples captured in urban streets, beaches, forests, and residential zones, ensuring broad applicability of the model.

- The team analyzed class imbalances and potential biases, especially in underrepresented waste types or scenes, which informed later steps in augmentation and training.

- Outliers and labeling inconsistencies were flagged through manual review and visualization dashboards.

This exploration phase laid the groundwork for more effective model training by exposing the dataset's limitations and guiding corrective strategies.

Augmentation Phase

To improve model generalization and mitigate overfitting, a comprehensive data augmentation pipeline was implemented using Roboflow. The augmentation strategy introduced controlled variability into the dataset, simulating real-world challenges in object appearance and environment.

Augmentation techniques included:

- Rotation: Randomized angles to account for different camera positions.
- Scaling and Resizing: To ensure robustness against varying object sizes.
- Brightness and Contrast Adjustments: Mimicking different lighting conditions across indoor and outdoor settings.
- Horizontal and Vertical Flips: Enhancing orientation tolerance.

Roboflow's augmentation tools allowed for streamlined and scalable transformation of the dataset. These enriched variations helped the segmentation and classification models to better recognize waste in diverse and unseen scenarios. Augmentation was applied selectively to maintain label integrity and preserve real-world plausibility.

This preprocessing phase significantly contributed to the improved performance of all downstream modules, including segmentation, classification, and estimation models.

4.2 Segmentation Module

Purpose and Role in the System

The segmentation module represents the **first stage in the system's multi-model architecture**, designed to localize and extract individual waste objects from input images. It acts as a **preprocessing step** for downstream models such as classification, volume estimation, and weight estimation. Accurate segmentation is critical, as the performance of subsequent models heavily depends on the quality and clarity of object boundaries generated at this stage.

Model Selection and Training Process

For object segmentation, the team adopted the **YOLOv11x** model, chosen for its real-time detection capabilities and strong performance in complex scenes. The model was fine-tuned using the **TACO (Trash Annotations in Context)** dataset, which includes a variety of waste categories in real-world environments.

The initial training run, using default parameters, yielded a baseline accuracy of **30%**. To enhance performance:

- **Bayesian Hyperparameter Optimization** was implemented to systematically search for optimal configurations, targeting:
    - Learning rate
    - Batch size
    - Anchor box dimensions

- **Advanced Data Augmentation** techniques were applied via **Roboflow**, introducing variations in:
    - Object orientation
    - Lighting conditions
    - Object scaling and flipping

Through iterative training and evaluation, the final accuracy improved to **57.3%**.

Additionally, the training process adopted the **SGD (Stochastic Gradient Descent)** optimizer, which contributed to improved convergence stability and better generalization on validation data.

Evaluation Metrics

To assess segmentation quality, the following metrics were computed:

- **mAP@0.5 = 64.1%**
- **Recall = 63.9%**
- **Precision = 56.2 %**

Performance analysis showed solid improvements in general detection accuracy, though certain challenges remained, particularly in scenes with overlapping waste or cluttered backgrounds. These were addressed through targeted dataset enhancement and retraining.

Tools and Technologies Used

- **Roboflow**: Dataset preprocessing, annotation handling, and augmentation.

- **PyTorch**: Core framework for model definition, training, and evaluation.

- **TensorBoard**: Training performance visualization and metric tracking.

Convex Hull Post-Processing Module

Purpose

After segmentation, some object masks were fragmented or incomplete due to occlusions or overlapping objects. The **Convex Hull Module** was introduced to post-process these outputs, aiming to reconstruct full object shapes and facilitate more accurate feature extraction for subsequent analysis stages (e.g., volume and weight estimation).

Methodology

Using **OpenCV**, the module applies a convex hull algorithm to each segmented contour:

1. **Contour Detection**: Using cv2.findContours() to extract object edges.

2. **Convex Hull Construction**: Using cv2.convexHull() to build the smallest convex polygon around each contour.

3. **Mask Refinement**: Creating smooth, filled shapes to correct fragmentation or gaps.

4. **Overlay Validation**: Visual comparisons between original images and reconstructed masks to verify alignment.

Tools and Libraries

- **OpenCV**: Primary tool for image processing and geometric reconstruction.
- **NumPy**: Supporting matrix operations and contour manipulation.
- **Matplotlib**: Used for visualization and qualitative performance validation.

4.3 Waste and Food Classification Module

Module Overview

The classification module forms the **second core component** of the system's multi-model architecture, following the segmentation stage. Its main objective is to **accurately identify and categorize waste and food items** from segmented images. This categorization supports downstream tasks such as volume estimation and environmental analytics.

Inspired by the GH2_MobileNet framework, this module implements a **hierarchical classification pipeline**, integrating:

- Instance segmentation (pre-detection),
- Primary object classification,
- Fine-grained food waste refinement.

The approach allows for the recognition of both general waste categories (e.g., plastic, metal, paper) and specific food types (e.g., vegetables, cakes, meat), ensuring semantic richness and analytical depth.

Backbone Architecture: MobileNetV2

To achieve high efficiency without sacrificing accuracy, the system leverages **MobileNetV2**, a lightweight convolutional neural network designed for resource-constrained environments such as mobile or edge devices.

Key architectural highlights:

- **Inverted Residual Blocks**: Efficiently link compressed layers, maintaining feature integrity while reducing parameters.
- **Linear Bottlenecks**: Prevent information loss by removing non-linearities from narrow layers.
- **Depthwise Separable Convolutions**: Decompose standard convolutions into depthwise and pointwise operations, significantly reducing computational load.

MobileNetV2 architecture includes:

- An initial convolutional layer with 32 filters,
- Followed by 19 residual bottleneck blocks,
- Each block includes expansion (1×1 ReLU6), depthwise (3×3), and projection (1×1 linear) layers.

This configuration provides a **favorable balance of speed, accuracy, and model compactness**, making it suitable for rapid inference in edge applications.

Multi-Stage Classification Workflow

Segmentation and Primary Classification

The process begins with the **Roboflow segmentation model**, deployed via hosted inference API. The model performs instance segmentation, returning:

- Object class predictions,
- Confidence scores,
- Masked regions (bounding boxes),
- Dynamically cropped sub-images.

This step handles a wide spectrum of waste materials including plastic containers, caps, straws, and organic waste. Each cropped object region is saved locally and indexed with metadata (e.g., timestamp, class, score).

Dynamic Cropping and Feature Extraction

Once objects are segmented, they are **dynamically cropped** and stored as standalone image samples. These cropped images:

- Are labeled with the primary class predicted,
- Are grouped by object type and detection confidence,
- Form the direct inputs to the **volume estimation model**,
- Serve as candidates for **secondary classification**, especially in the case of food waste.

This intermediate step bridges segmentation with fine-grained classification and estimation logic.

Refined Food Waste Classification

Given the heterogeneous nature of food waste, a **second-level classifier** is employed to enhance specificity. A **dedicated Roboflow API**, trained on curated food categories (e.g., meat, fruits, desserts), is triggered for all instances initially labeled as “food”.

This two-tiered refinement ensures **granular food classification**, boosting the reliability of downstream **volume estimation algorithms** that rely on food type-specific density profiles.

Tools and Technologies

- **MobileNetV2 (TensorFlow/PyTorch)**: Efficient CNN backbone for classification.
- **Roboflow Hosted Inference APIs**: For segmentation and food classification workflows.

- **OpenCV / NumPy / Pandas**: For image manipulation, cropping, and data logging.

4.4 Volume Estimation Module

Module Objective

The Volume Estimation Module is designed to quantify the physical volume of segmented food items using only single-view RGB images. This capability is essential for food waste monitoring, consumption analytics, and sustainable behavior modeling. The approach integrates Mask R-CNN-based segmentation with monocular depth estimation, allowing accurate 3D volume inference without requiring specialized hardware.

Core Pipeline Architecture

Instance Segmentation using Mask R-CNN

The process begins with precise instance segmentation using a Mask R-CNN architecture fine-tuned on the UNIMIB2016 dataset — a dataset specifically designed for food recognition in real-world conditions.

- **Model Features:**
    - Region Proposal Network (RPN) for object localization,
    - Pixel-level binary masks for object boundaries,
    - High performance under occlusion, mixed ingredients, and variable lighting conditions.
- **Training Strategy:**
    - Transfer learning from a COCO-pretrained Mask R-CNN backbone,
    - Fine-tuning on UNIMIB2016 with augmentations (e.g., color jitter, rotation, scaling),

- Validation against manual segmentation annotations.

Each segmented food item is converted into a binary mask, from which its projected area is computed. To map image pixels to physical dimensions, the system references real-world scale anchors within the image (e.g., dishware or utensil sizes).

Depth Estimation and 2.5D Projection

To reconstruct the object's geometry from a single RGB image:

- A pretrained monocular depth estimation model (e.g., MiDaS or DPT) is used to infer per-pixel depth across the segmented mask.
- The segmentation mask is overlaid with the depth map to create a height map or voxelized 2.5D model.

This allows estimation of:

- Object surface curvature,
- Volume enclosed within the reconstructed shape,
- Projection of depth values into real-world units via camera calibration and reference scaling.

Volume Computation

Once the depth-aligned segmentation mask is constructed:

- A discrete volume integration step computes the object volume by summing the pixel-wise products of area and depth.
- The final output is returned in milliliters (mL), normalized by pixel-to-metric scale derived from reference objects in the scene.

This pipeline eliminates the need for depth sensors or stereo cameras, making it ideal for mobile or low-resource deployment.

Dataset and Optimization

UNIMIB2016 Dataset

The UNIMIB2016 dataset was used to adapt the model to food-specific visual domains. Key advantages:

- High intra-class variance,
- Annotated food types with occlusions and overlapping elements,
- Real-world lighting conditions.

Optimization Strategy

- Initial training used the COCO-pretrained Mask R-CNN weights.
- Continued fine-tuning on UNIMIB2016 with task-specific augmentations.
- Augmentation types: random brightness, affine transformations, zoom, and angle rotation.

This strategy maximized generalization, enabling the model to handle complex meal types and lighting setups.

Evaluation Metrics and Performance

- Segmentation Metrics:
    - IoU (Intersection over Union): Evaluates mask overlap with ground truth,
    - Precision and Recall: For detection consistency.
- Volume Accuracy:
    - Mean Absolute Percentage Error (MAPE): Below 12% for core food types (e.g., pasta, rice, fruit),

- Benchmarking performed against ground-truth volumes obtained via water displacement experiments.

Limitations and Future Work

While highly effective for homogeneous and clearly visible food items, the model may show reduced accuracy for:

- Mixed dishes with overlapping components,
- Items within transparent or reflective containers.

Potential enhancements include:

- Multi-view stereo capture for richer 3D reconstruction,
- Integration of low-cost ToF sensors,
- Probabilistic models for uncertainty quantification in volume estimates.

Tools and Libraries

- Mask R-CNN (Detectron2/PyTorch): Backbone for instance segmentation,
- Monocular Depth Models (e.g., MiDaS): For pseudo-3D reconstruction,
- OpenCV + NumPy: For volume integration and scaling.

4.5 Weight Estimation

Weight estimation is a foundational step in the assessment of organic and non-organic waste for hydrogen extraction applications. In the absence of direct weight measurements, the adopted approach in this system relies on a physics-informed estimation strategy grounded in volumetric analysis and material-specific density mapping. This method leverages the basic physical relationship:

$$W = \rho \cdot V \cdot \phi$$

- W: Estimated weight (in grams)
- ρ: Material density (in g/cm^3), retrieved based on the classified object category
- V : Estimated object volume (in cm^3), derived from instance segmentation and depth approximation
- ϕ: Fill ratio, a scalar representing the proportion of the object's internal volume that is occupied (accounts for hollow or porous materials)

Volume Acquisition

We initiate the pipeline by estimating the volume V of each waste object using a combination of instance segmentation (YOLO-based) and monocular depth prediction models. Integrating these models enables the derivation of object boundaries and depth cues from a single RGB input, allowing volume calculation via reconstructed voxel grids or bounding-box integration. This process is computationally efficient and preserves spatial integrity without requiring stereo vision setups or LiDAR [34].

Density Determination

The next critical step is the determination of density values for each material class. Unlike volume estimation, density cannot be computed algorithmically or inferred from visual features alone. It is inherently a material property and thus requires empirical data and domain-specific knowledge.

Fill Ratio

The **fill ratio ϕ** is introduced as a correction factor to account for the object's internal structure (e.g., solid vs. hollow). This parameter is predefined per object class based on domain-specific knowledge or physical measurement studies

Recognizing this, we conducted a **comprehensive literature review and empirical synthesis** to compile accurate and representative density values for a wide spectrum of waste categories. This step was particularly essential for mixed or organic waste types (e.g., food residues, paper, cardboard, PET plastics, aluminum foil, etc.) where density varies significantly depending on composition, state (wet/dry), and compaction level [35].

The reviewed studies included physical characterization data from municipal waste audits, laboratory testing, and recycling facility reports. This process yielded a refined list of over 40 waste categories, each with its associated density range and median value. These values were encoded into a structured **JSON schema**, enabling:

- Easy lookup during real-time estimation
- Support for updates based on new data
- Hybrid interpolation for ambiguous or composite materials

Examples of adopted densities include:

- **Beef**: 1.06 g/cm^3
- **Pie**: 0.88 g/cm^3
- **Plastic Bottle (Clear)**: 1.38 g/cm^3
- **Cardboard (Dry)**: 0.68 g/cm^3
- **Cooked Organic Food Waste (Average)**: 1.01 g/cm^3
- **Apple Cores**: 0.81 g/cm^3
- **Chicken Bones (Dry)**: 1.55 g/cm^3
- **Polyethylene Terephthalate (PET)**: 1.38 g/cm^3
- **High-Density Polyethylene (HDPE)**: 0.95 g/cm^3

These values were extracted or approximated from standardized datasets and experimental studies [36]. When multiple sources reported varying values for the same material, we computed a weighted average based on environmental conditions (e.g., moisture content, compaction), ensuring contextual relevance to urban waste streams.

JSON Schema Design for Density Integration

To operationalize density mapping, we embedded the material-density pairs into a dedicated JSON file, serving as a lightweight, queryable database during system runtime. The JSON format was selected for its:

- Language-agnostic integration across Python-based modules
- Readability and updatability without system recompilation
- Support for nested representations, allowing the mapping of composite items (e.g., milk cartons with paper/plastic layers)

This modular design ensures long-term scalability, particularly when expanding the object detection classes or adapting the system to new regional waste compositions. Density values are queried per detected class label and used directly in the weight equation.

Validation and Accuracy Considerations

Although this method cannot fully replicate the accuracy of physical weighing, it achieves high fidelity for homogeneous and well-defined materials. Errors primarily stem from:

- Incorrect object classification (e.g., mistaking PET for PVC)
- Deviations in object posture or fragmentation
- Density variance within the same material class

To mitigate these, future improvements may incorporate probabilistic ranges or confidence intervals in the density lookup process, similar to Bayesian updates as suggested in waste analytics literature [37].

4.6 Hydrogen Ratio Estimation in Organic Waste

Hydrogen content within waste materials is a critical factor when evaluating the energy potential of organic waste for sustainable hydrogen production. Unlike traditional material characterization techniques, estimating the hydrogen ratio in heterogeneous waste requires both chemical composition analysis and empirical data derived from various material classes. Our system models hydrogen content dynamically, using a robust process that accounts for material-specific hydrogen-to-carbon ratios, moisture content, and chemical bonds.

Hydrogen Content Determination: Methodology

Estimating the hydrogen content in waste materials is a fundamental step in assessing their potential for green hydrogen production. Since direct chemical analysis is not feasible in real-time systems, our approach relies on a computational estimation strategy that combines pre-estimated object weight and material-specific hydrogen mass fractions.

Once the object's **weight** W(in grams) is calculated using volumetric and density-based modeling, the total extractable hydrogen is estimated using the following formula:

$$H_{corr} = W \cdot H_{frac}$$

The calculation was refined further by factoring in the **moisture content (M)**, as water significantly impacts hydrogen yields due to its inherent hydrogen atoms.

For organic materials (e.g., food waste, cardboard, and agricultural residues), **moisture correction** is essential to account for water weight, which would otherwise distort hydrogen calculations. The corrected hydrogen content H_{corr} is computed as:

$$H_{corr} = W \cdot (1 - M) \cdot H_{frac}$$

Where M represents the moisture content, W is the object weight, and $H_{frac}$is the hydrogen mass fraction.

Extensive Research and Data Collection on Hydrogen Ratios

Our team undertook an **extensive literature review** to determine the most accurate and up-to-date hydrogen ratios for the wide range of organic waste types encountered in municipal streams. These ratios were compiled into the system's **JSON file**, ensuring that each material's specific hydrogen content could be referenced and updated as new studies became available.

The reviewed research spans various material categories, with representative hydrogen mass fractions summarized below:

- **Food waste**: The hydrogen content in food waste is generally high, with typical hydrogen fractions ranging from 5.5 to 6.5 wt%. This range is influenced by the organic composition of the waste (e.g., fruits, vegetables, meats, and fats) [38].
- **Paper/Cardboard**: Paper and cardboard materials show hydrogen content between 4.5 and 5.5 wt% on a dry basis, depending on their lignocellulosic structure and moisture levels [39].
- **Plastics (e.g., PET, HDPE)**: Plastics generally have lower hydrogen content than organic materials, with values ranging from 3.5 to 4.5 wt%, as their polymeric chains contain fewer hydrogen atoms relative to carbon and oxygen [40].

Accurate hydrogen content estimation plays a pivotal role in optimizing the financial sustainability of waste-based hydrogen production. Our system's ability to dynamically adjust hydrogen ratios, coupled with real-time data updates via the JSON schema, enables scalable and adaptable waste management solutions. Future research will continue to refine hydrogen ratio predictions by incorporating advanced material decomposition techniques and larger-scale waste audits.

4.7 Waste Cost Calculation Based on Hydrogen Content

The pricing model for organic waste is primarily driven by its **hydrogen content**, which plays a crucial role in determining its value for **biohydrogen production**. The system uses a **dynamic pricing mechanism** for customers selling their waste, but also incorporates larger-scale transactions with companies specializing in biohydrogen production.

Principles of Cost Calculation

The cost for the organic waste is calculated based on the **hydrogen content** H_{corr}, reflecting the potential for hydrogen extraction. The cost equation is:

$$Cost_{object} = H_{corr} \times P_{per\ gram} \times Share\ Factor$$

Where:

- H_{corr}: the hydrogen content in grams, calculated based on the weight and hydrogen fraction of each material, adjusted for moisture content.
- $P_{per\ gram}$: the price per gram of hydrogen, based on market conditions.
- ***Share Factor***: 40% of the total hydrogen value is allocated to the customer, while the remaining 60% is retained by the company for processing and operational costs.

This approach ensures that customers are compensated for the hydrogen content in their waste, while the company can ensure profitability for the waste collection and conversion process.

Market Considerations and Hydrogen Pricing

The price of hydrogen in the market typically fluctuates between **$3 to $6 per kilogram** [41]. The price per gram of hydrogen is derived from this range, giving us the following:

- **Lower bound price:** 3 $ per kg÷1000=0.003 $ per gram
- **Upper bound price:** 6 $ per kg÷1000=0.006 $ per gram

For transactions with individual customers, the pricing is based on this model. However, when dealing with larger companies, the pricing dynamics shift.

Example Calculation

Step 1: Define Properties of the Waste

Assume we have the following properties for the waste (e.g., food waste):

- The weight of the object: 5000 grams (5 kilograms)
- The hydrogen fraction H_{frac} for food waste: 0.06 (6% hydrogen by weight)
- The moisture content M: 0.02 (2% moisture content)

Step 2: Calculate Hydrogen Content H_{corr}

First, we calculate the hydrogen content H_{corr}:

$$H_{corr} = 5000 \text{ grams} \times (1-0.02) \times 0.06 = 5000 \times 0.8 \times 0.06 = 240 \text{ grams of hydrogen}$$

Step 3: Calculate the Customer's Share of the Hydrogen Price

Now, we will calculate the cost based on the market price for hydrogen, using both the lower and upper bounds of the price per gram of hydrogen (as per the market price range from $3 to $6 per kilogram, i.e., $0.003 to $0.006 per gram).

Lower Bound Price Calculation

If the market price is at the lower bound ($0.003 per gram), the customer would receive:

$$Cost_{object} = 240 \text{ grams} \times 0.003 \text{ \$ per gram} \times 0.4 = 0.288 \text{ \$}$$

Upper Bound Price Calculation

If the market price is at the upper bound ($0.006 per gram), the customer would receive:

$$Cost_{object} = 240 \text{ grams} \times 0.006 \text{ \$ per gram} \times 0.4 = 0.576 \text{ \$}$$

Step 4: Convert the Price to Egyptian Pounds (EGP)

Using an exchange rate of 50 EGP per USD, we can convert the prices to Egyptian pounds:

- **Lower bound price in EGP**:

$$0.288\ \$ \times 50\ \text{EGP}/\$ = 14.4\ \text{EGP}$$

- **Upper bound price in EGP**:

$$0.576\ \$ \times 50\ \text{EGP}/\$ = 28.8\ \text{EGP}$$

So the price will be in the range from (150 EGP to 300 EGP for a kilogram of hydrogen), and we preferred to work at the average price of 4.5 $ ≈ 225 EGP for a kilogram of hydrogen

Transactions with Large Companies

In addition to selling organic waste to individual customers, our system will also engage in **larger-scale transactions with companies** that specialize in **green hydrogen production**. These companies will purchase the organic waste in bulk for conversion into biohydrogen. Due to the scale and complexity of these transactions, the pricing structure will be **contractual** rather than dynamic, as it is with individual customers.

Contractual Pricing

The pricing for large companies will be based on **tons of organic waste** rather than individual units. These transactions will involve long-term contracts that establish fixed pricing for the organic waste, considering factors such as:

- **Volume of waste** purchased (typically measured in tons).
- **Consistency and quality** of the organic waste provided.
- **Hydrogen yield potential** based on the material composition.
- **Market conditions** for biohydrogen and the extraction efficiency of the technology used by the companies.

These contracts will involve negotiations to determine the price per ton of waste. The price will be reflective of the market value of hydrogen, but adjusted for factors like the long-term nature of the agreement and the buyer's operational costs.

Example Calculation

For a typical transaction with an individual customer, the cost calculation would proceed as outlined in the previous section. However, for a **large company** purchasing waste in bulk (let's say 50 tons), the pricing would be based on a **pre-agreed contract price**. For example:

- If a contract specifies a **price of $100 per ton of waste**, the total cost for 50 tons would be:

$$\text{Total Cost} = 50 \text{ tons} \times 100 \text{ \$ per ton} = 5000 \text{ \$}$$

This pricing would not fluctuate based on the hydrogen content or market price of hydrogen in the same way that customer transactions do. Instead, the contract price would remain consistent throughout the term of the agreement, unless renegotiated based on external factors like production costs or market shifts.

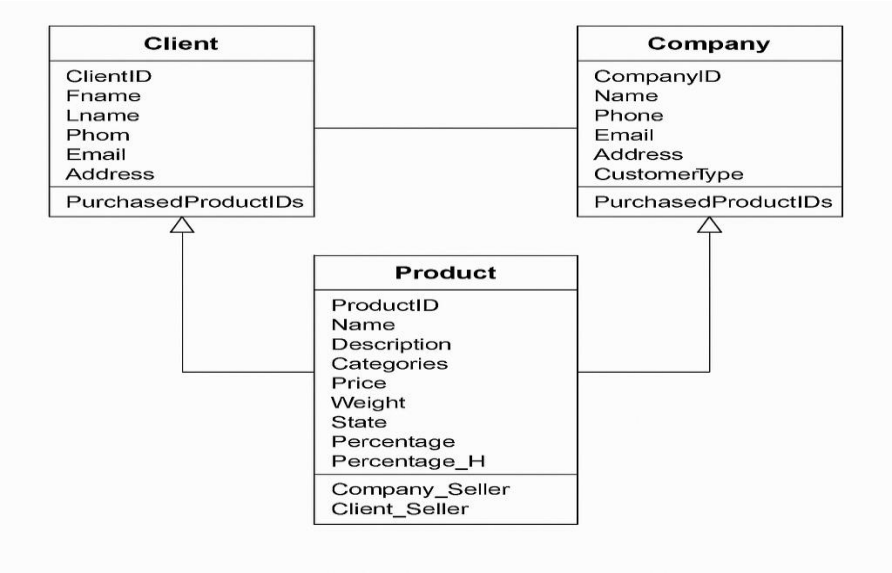
Conclusion

The waste pricing system accounts for both **individual transactions** with customers and **larger-scale contracts** with companies. For customers, the pricing is based on hydrogen content and market fluctuations, ensuring fair compensation based on the material's value. For larger companies, the pricing will be based on long-term contracts, which allow for stable pricing based on bulk waste purchases. This dual approach ensures flexibility and fairness for all parties involved, from individual waste generators to large-scale industrial buyers.

4.8 Database Technology

In this project, we used MongoDB, a NoSQL document-oriented database. MongoDB stores data in flexible, JSON-like documents, which allows for a dynamic schema and efficient handling of complex, nested data structures.

Figure (9) UML Design Overview



The UML diagram was designed to reflect the structure and relationships between the main collections: Client, Company, and Product.

- The Product document references the seller only (either a client or a company).

- Buyers are not directly stored in the Product collection. Instead, the PurchasedProductIDs field is included in the Client and Company collections if their type includes "Buyer".

- The CustomerType field in both Client and Company documents indicates whether the entity is a Buyer, Seller, or Both.

Collection: Client

Structure:

- ClientID: Unique identifier for the client.

- Fname: First name.

- Lname: Last name.

- Phone: Client’s phone number.

- Email: Client’s email address.

- Address: Physical address.

- CustomerType: Indicates whether the client is a Buyer, Seller, or Both.

- PurchasedProductIDs: List of product IDs the client has purchased (only present if the client is a Buyer or Both).

Example:


```
{
  "ClientID": "C001",
  "Fname": "John",
  "Lname": "Doe",
  "Phone": "+1234567890",
  "Email": "john.doe@example.com",
  "Address": "123 Main St, Cityville",
  "roles": {"buyer":1,"seller":"1"},
  "PurchasedProductIDs": ["P101", "P205"]
}
```


Collection: Company

Structure:

- CompanyID: Unique identifier for the company.

- Name: Company name.

- Phone: Company’s phone number.

- Email: Company’s email address.

- Address: Physical address.

- Domain: The field or domain in which the company operates.

- CustomerType: Indicates whether the company is a Buyer, Seller, or Both.

- PurchasedProductIDs: List of product IDs the company has purchased (only if Buyer or Both).

Example:


```
{
  "CompanyID": "COM01",
  "Name": "Tech Solutions Ltd.",
  "Phone": "+987654321",
  "Email": "info@techsolutions.com",
  "Address": "456 Business Rd, Technopolis",
  "Domain": "Technology",
  "roles": {"buyer":1,"seller":"0"},
  "PurchasedProductIDs": ["P101", "P202", "P303"]
}
```


Collection: Product

Structure:

- ProductID: Unique identifier for the product.

- Name: Product name.

- Description: Brief description of the product.

- Categories: Array of product categories.

- Price: Product price.

- Weight: Product weight.

- State: Product condition or availability.

- Percentage: Discount percentage or other business metric.

- Percentage_H: Historical percentage or comparative metric.

- Company_Seller: ID of the selling company (if applicable).

- Client_Seller: ID of the selling client (if applicable).

Example:


```
{
  "ProductID": "P101",
  "Name": "Smartphone X",
  "Description": "Latest model with high performance features",
  "Categories": ["Electronics", "Mobile"],
  "Price": 699.99,
  "Weight": "200g",
  "State": "New",
  "Percentage": 10,
  "Percentage_H": 8,
  "Seller": "COM01"
}
```

4.9 Website Implementation

Frontend Architecture (React.js)

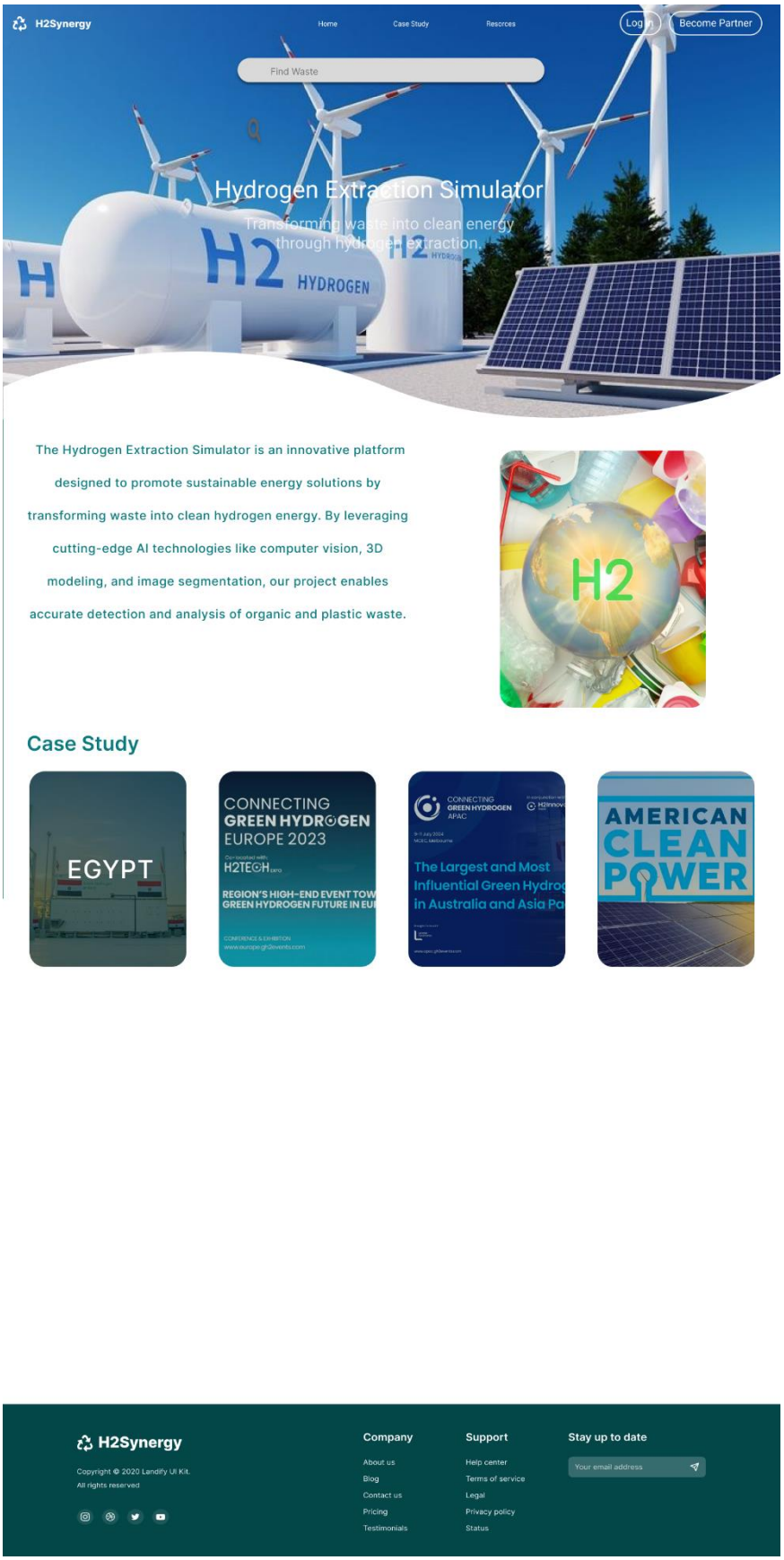
The frontend is implemented using React.js, a component-based JavaScript library. The architecture is modular, scalable, and responsive, ensuring optimal performance on both desktop and mobile devices.

Key Components:
- Navbar & Footer: Consistent layout elements.
- Home Page: Introduction to the platform and its benefits.
- Authentication: LogIn Form, Register Form (Client & Company), Password Recovery.
- User Dashboards: Client Account, Company Account, Profile Pages.
- Waste Management: Upload Waste, Products, Waste Products.
- AI Results View: Processed data output shown to users.
- Wish List: Allows users to track desired outcomes or rewards.

UI/UX Design

The user interface follows a clean and intuitive layout. High-fidelity designs guide the layout of components, ensuring a smooth user experience. Users can easily navigate the site, upload images, and view results.

Screens include:
- Home Page
- Register / Login / Recovery Forms
- Client & Company Accounts
- Waste Upload and Results Pages
- AI Visualization Panel
- Navigation: Top nav and collapsible side nav


H2Synergy
Home
Case Study
Resorces
Log in
Become Partner
Find Waste
Hydrogen Extraction Simulator
Transforming waste into clean energy through hydrogen extraction.
H2
HYDROGEN
The Hydrogen Extraction Simulator is an innovative platform designed to promote sustainable energy solutions by transforming waste into clean hydrogen energy. By leveraging cutting-edge AI technologies like computer vision, 3D modeling, and image segmentation, our project enables accurate detection and analysis of organic and plastic waste.
H2
Case Study
EGYPT
CONNECTING GREEN HYDROGEN EUROPE 2023
H2TECH
REGION'S HIGH-END EVENT TOW
GREEN HYDROGEN FUTURE IN EU
CONNECTING GREEN HYDROGEN APAC
The Largest and Most Influential Green Hydrog
in Australia and Asia Pa
AMERICAN CLEAN POWER
H2Synergy
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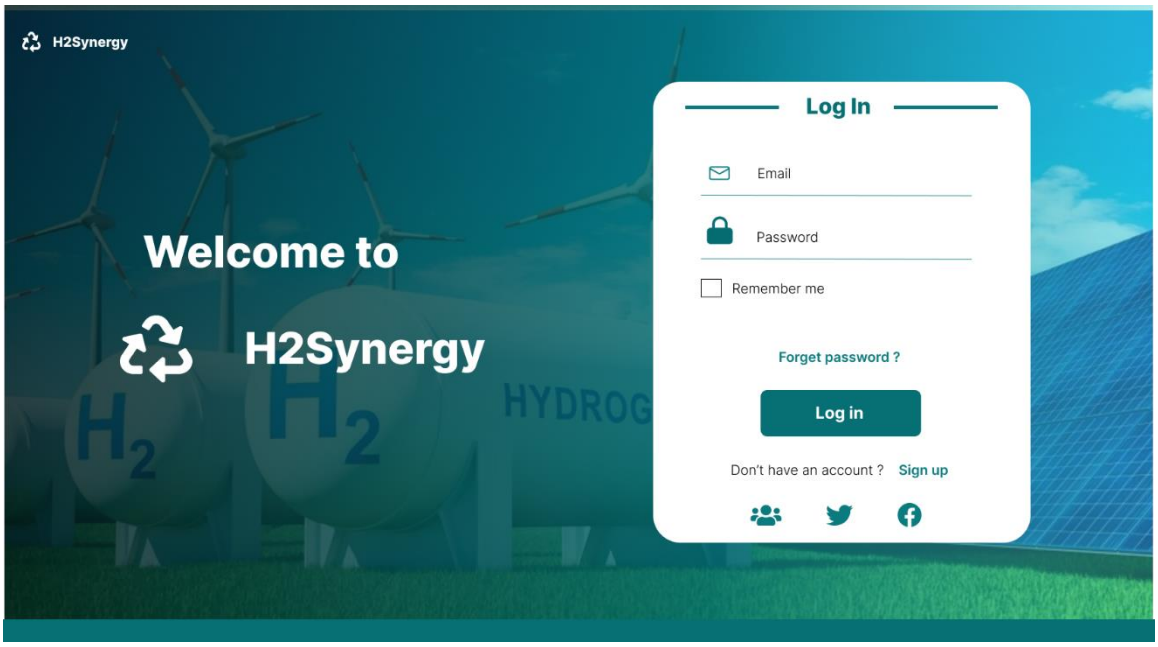
Company
About us
Blog
Contact us
Pricing
Testimonials
Support
Help center
Terms of service
Legal
Privacy policy
Status
Stay up to date
Your email address

1. Home Page

The Home Page serves as the main entry point to the platform. It introduces users to the project's mission — producing green hydrogen from organic waste — and outlines the environmental benefits. The layout is clean and intuitive, with clear call-to-action buttons that guide users to register, log in, or explore further.

- **<u>Home Page – Components</u>**
  - **Navbar**: Navigation bar with links to
    - Home, Login, Register.
  - **Hero Section**: Introduction text +
    - CTA button.
  - **Feature Highlights**: Cards showing
    - benefits (Eco-friendly, AI-powered,
    - etc.).
  - **Footer**: Platform info, contact, and
    - legal links.

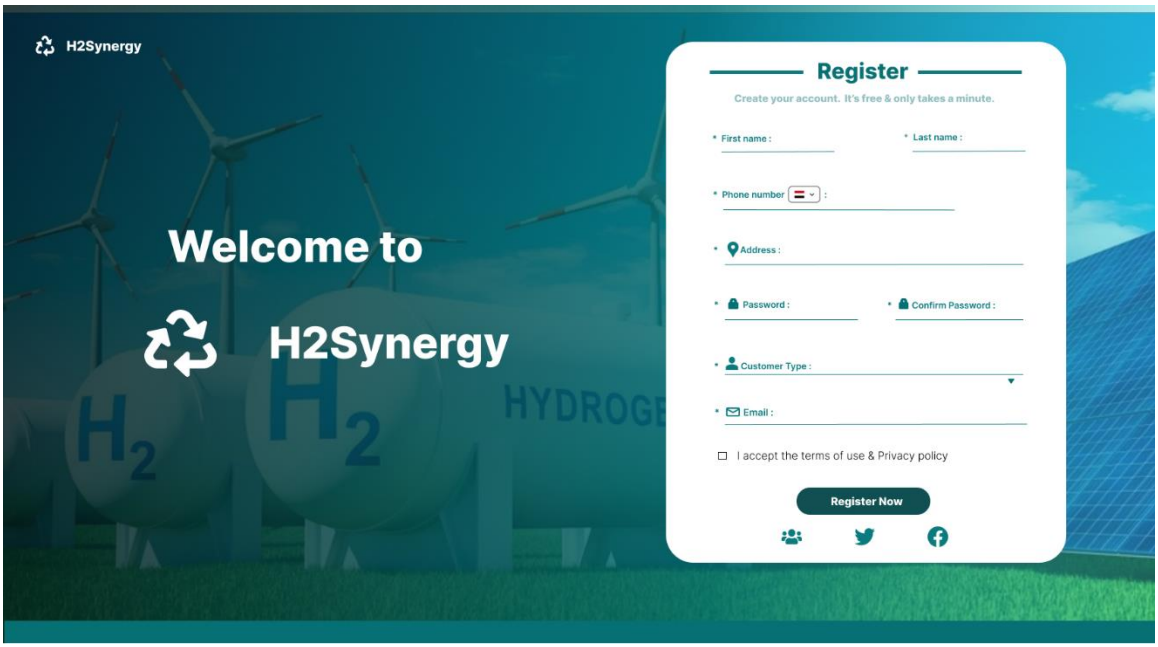
2. Log In Form



The login form provides a secure interface for returning users to access their accounts. It includes validation features for email and password fields, along with a link to the password recovery page. The design prioritizes simplicity and user-friendliness to streamline access.

- **Login Form – Components**
  - **Email Input**: Validated email field.
  - **Password Input**: Secure input with show/hide toggle.
  - **Login Button**: Auth trigger.
  - **Forgot Password Link**: Redirection to recovery.
  - **Error Message Area**: Conditional feedback UI.

3. Client Register Form

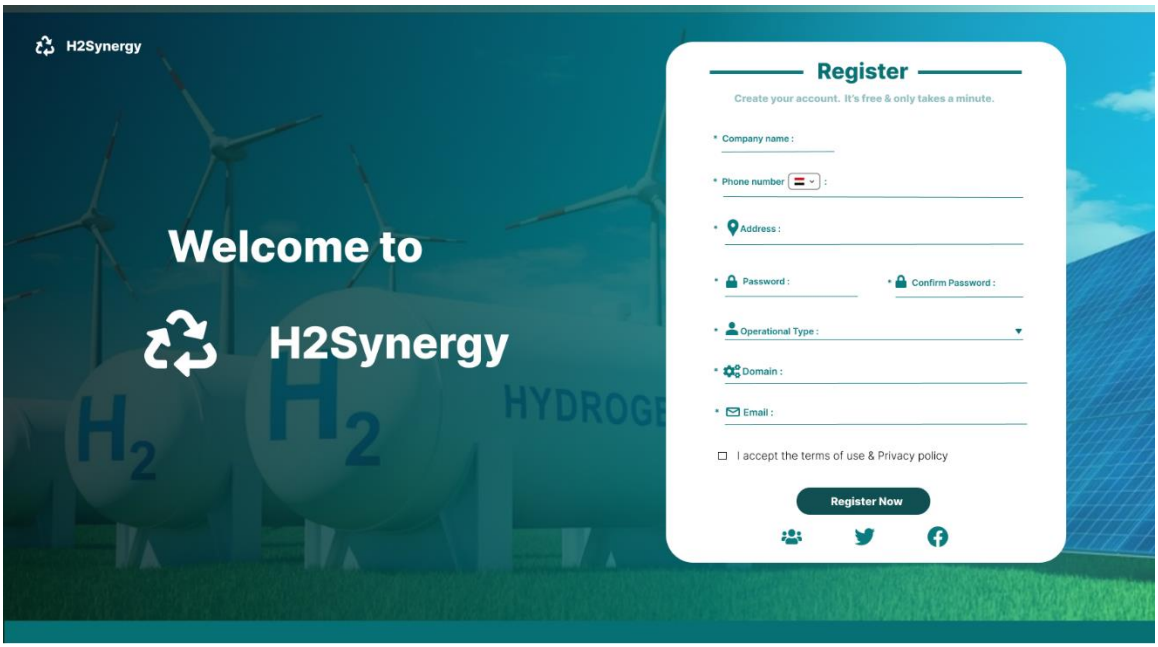


This form is designed for individual users to create accounts. It collects essential information such as name, email, password, and optional profile details. The interface is minimal and focused on reducing friction during the registration process.

- **Client Register Form – Components**
  - **Name, Email, Password Inputs**
  - **Confirm Password Field**
  - **Register Button**
  - **Validation & Feedback Logic**

---

4. Company Register Form



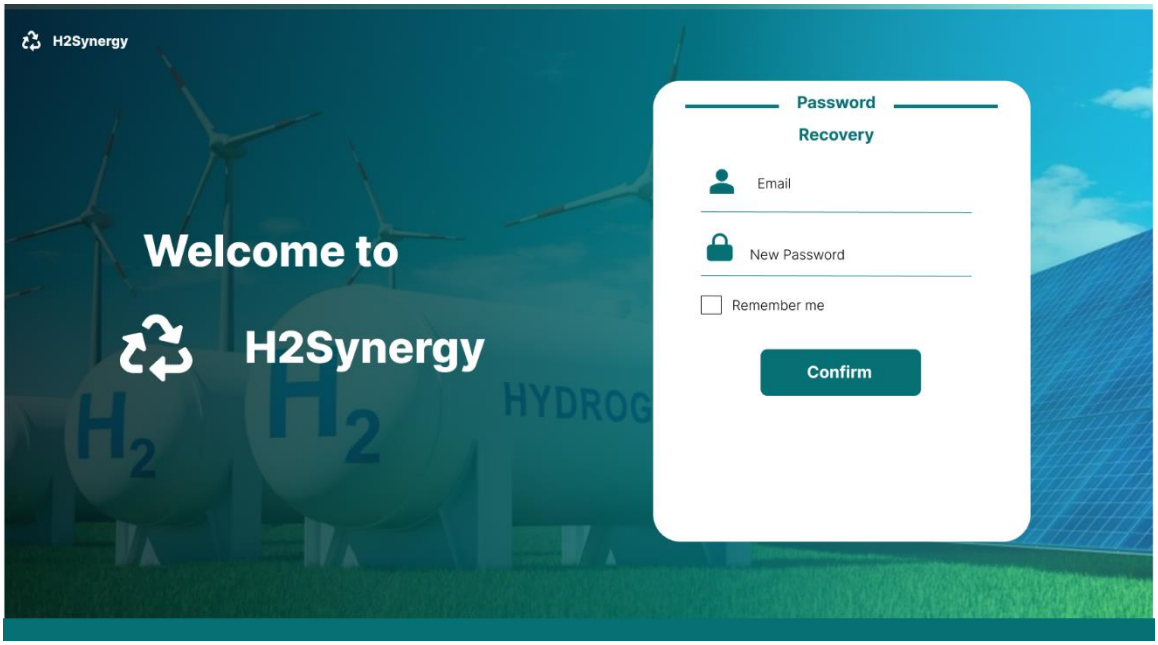
Tailored for organizations and businesses, this form includes additional fields to capture company details such as organization name, business type, and contact person. It ensures that companies can be onboarded efficiently while maintaining data accuracy.

- **Company Register Form – Components**
  - **Company Name, Type**
  - **Contact Person, Email**
  - **Phone Input**
  - **Submit Button**

5. Password Recovery

If users forget their credentials, this interface allows them to request a secure password reset link via email. It ensures continued access without compromising security.

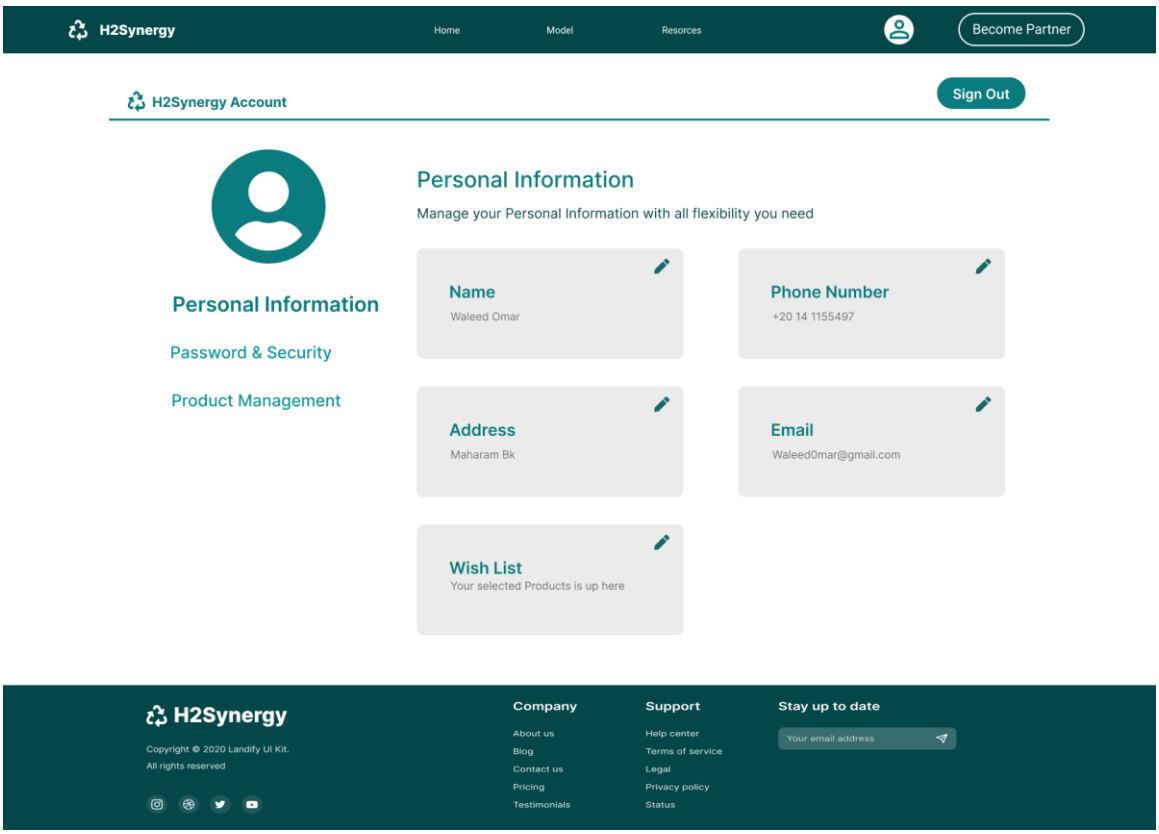
- **Password Recovery – Components**
  - **Email Input**
  - **Send Button**
  - **Status Feedback**

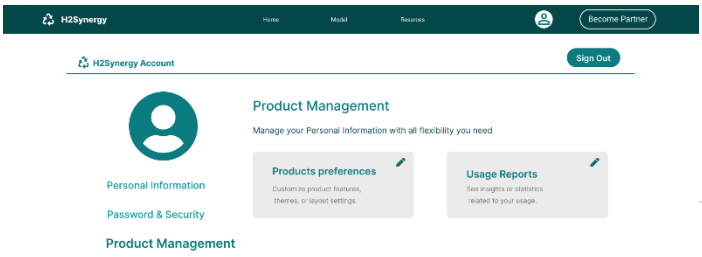


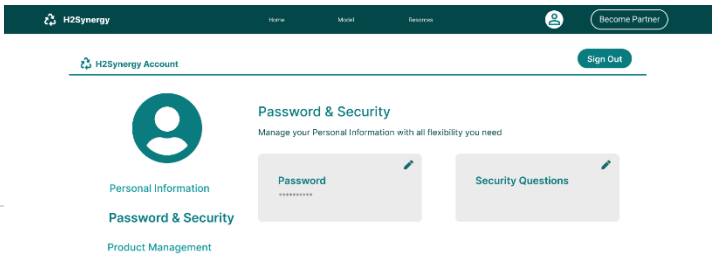
6. Client Account

The Client Account dashboard allows individual users to manage their waste submissions, view hydrogen yield results, and track the status of ongoing analyses. It provides a personalized experience, including visual indicators for past submissions and expected rewards.

- **<u>Client Account – Components</u>**
  - **Dashboard Cards**: Hydrogen output, previous uploads.
  - **History Table**
  - **Profile Quick View**



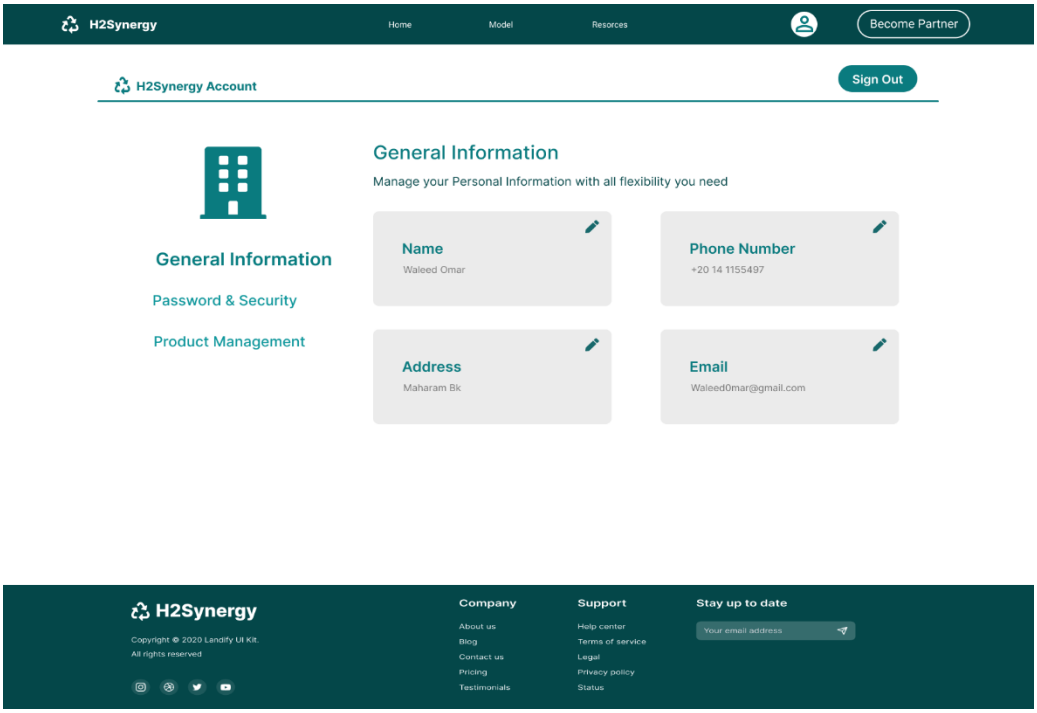


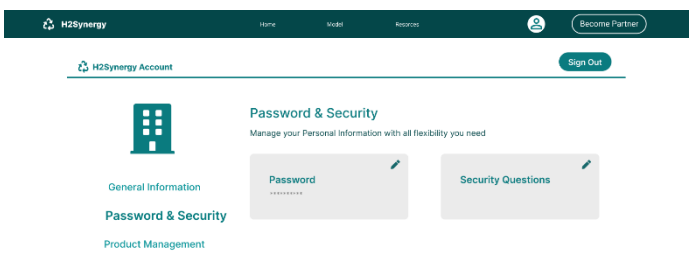


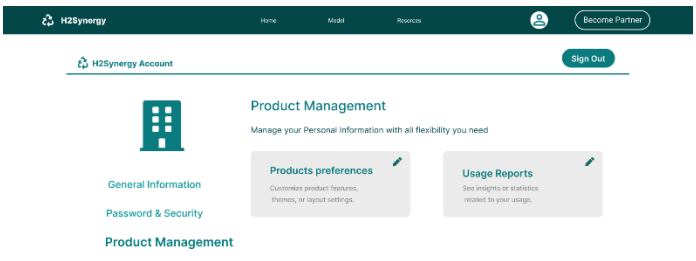
7. Company Account

Similar to the Client Account, this dashboard is designed for businesses with features tailored to larger-scale waste submissions. It includes batch upload tracking, submission analytics, and hydrogen production estimates for each batch.

- **Company Account – Components**
  - ➢ **Batch Submission Viewer**
  - ➢ **Analytics Cards**
  - ➢ **Sidebar Menu**



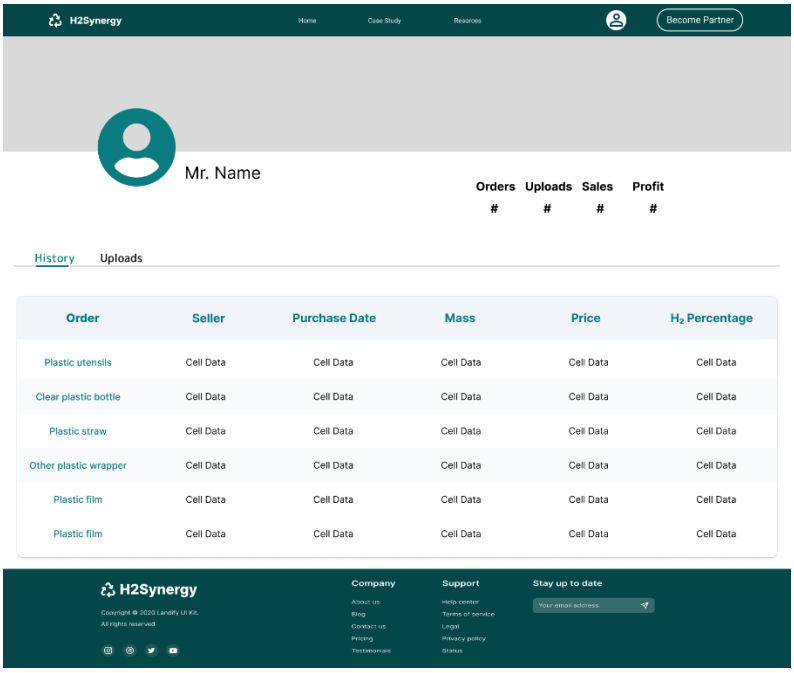


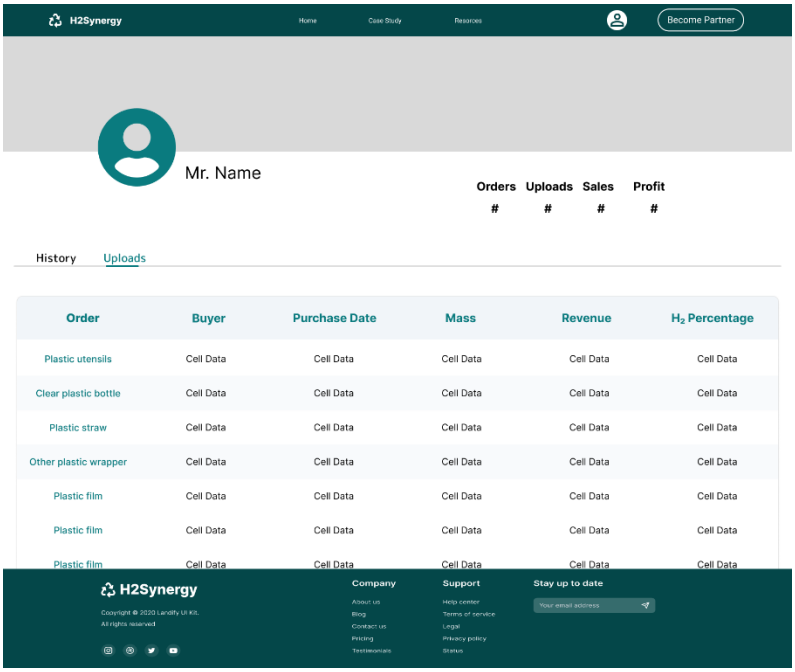


8. Profile Page

This page enables users to manage their personal information, including updating profile details, changing passwords, and accessing their upload history. It contributes to personalization and enhances user control over data.

- **Profile Page – Components**
  - **Form Fields** (name, email, password)
  - **Save Button**
  - **History Log**

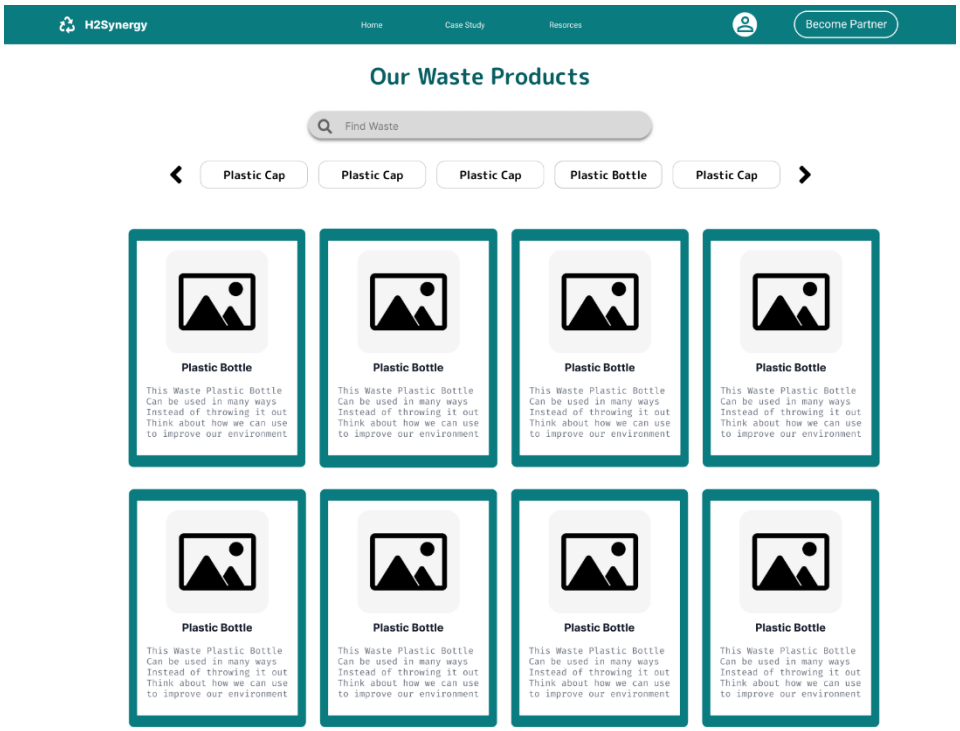


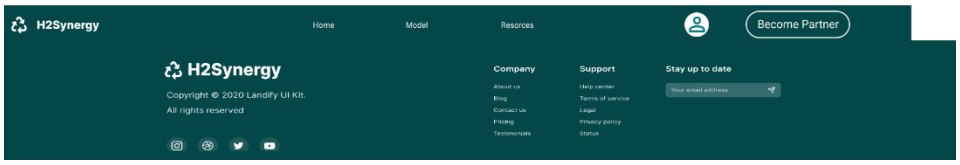


9. Wish List

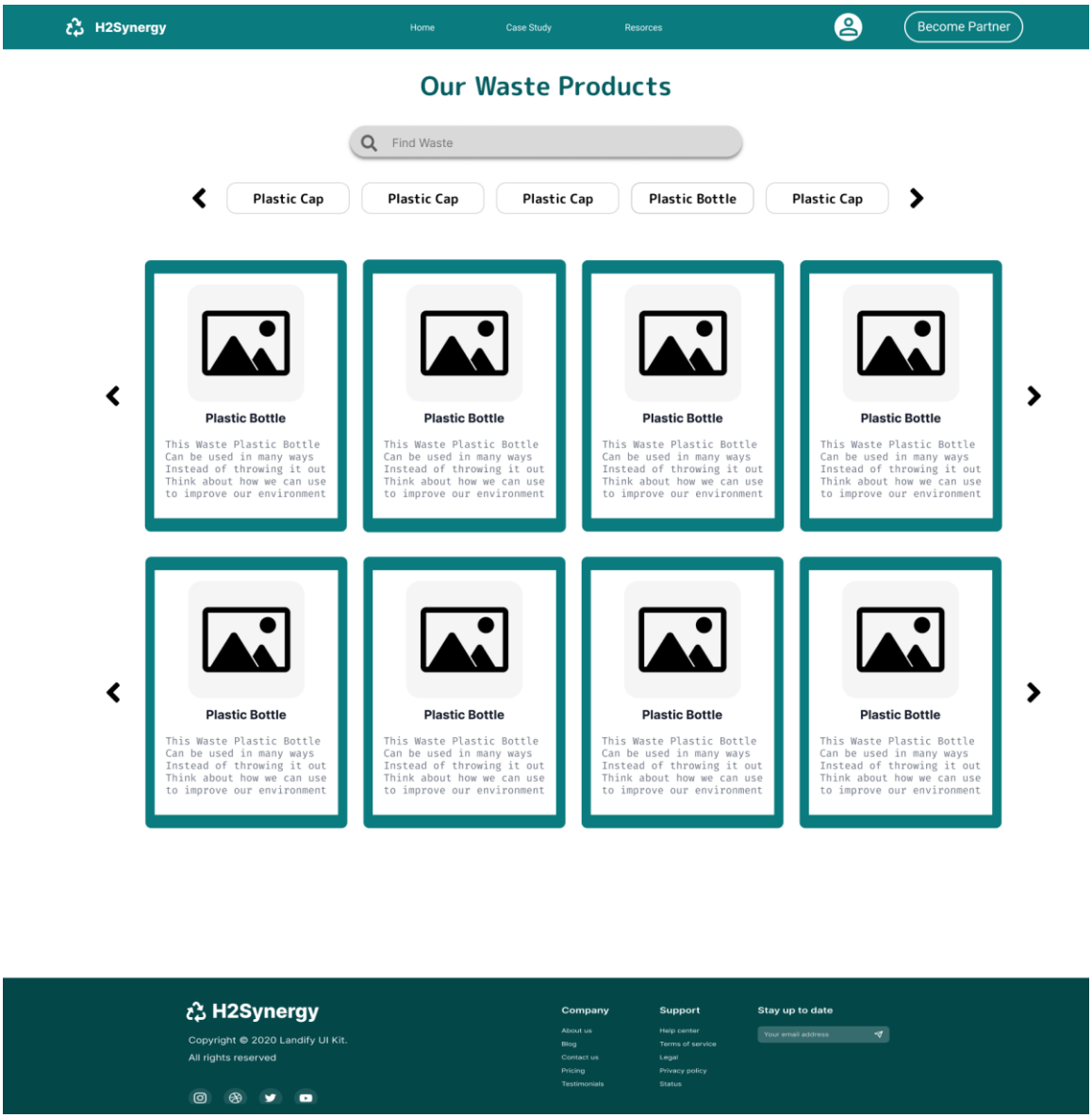
The Wish List is a motivational feature that lets users track potential incentives or future platform features they are interested in. It supports engagement and long-term participation by visualizing possible benefits tied to sustained contributions.

- **Wish List – Components**
  - **Incentive List Cards**
  - **Add to Wishlist Button**
  - **Progress Bar**





10. Products Page



This screen showcases the output products or categories generated from waste processing. It may include examples of energy products, fertilizers, or hydrogen-related outputs — helping users visualize impact and value creation.

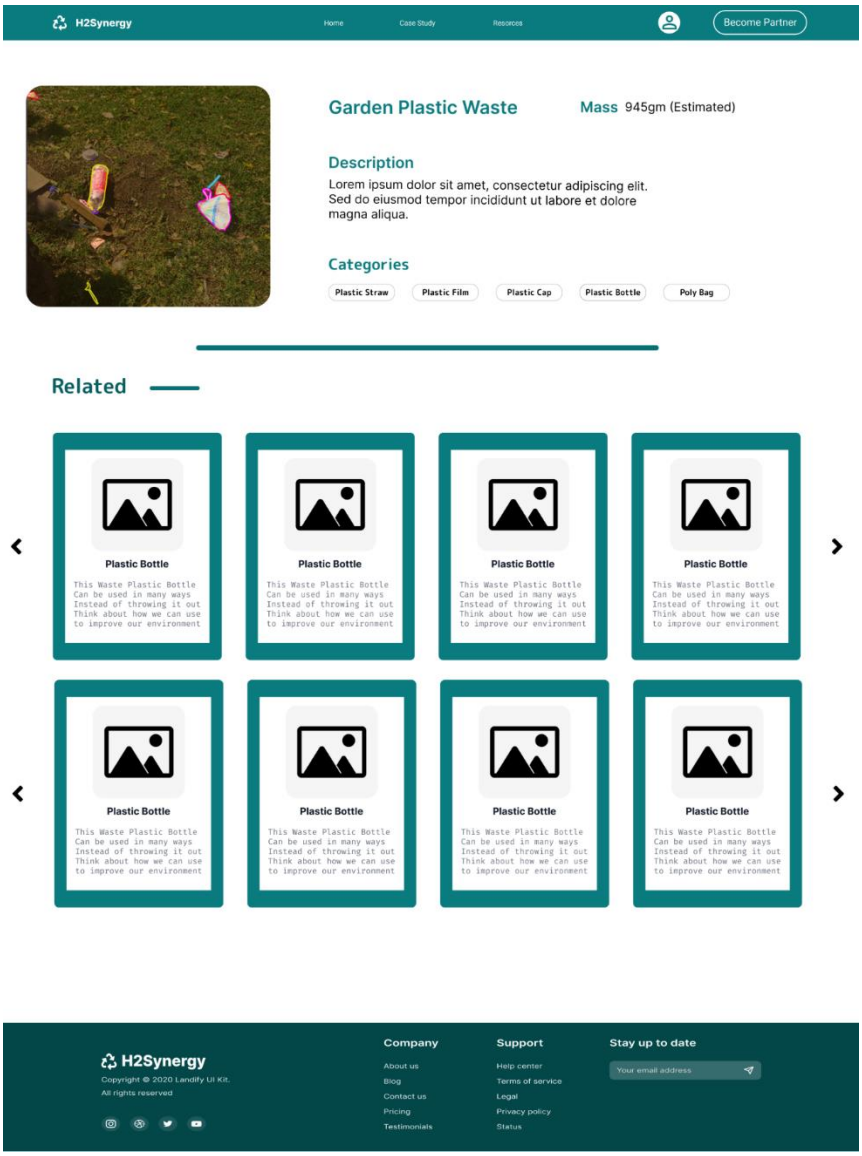
- **Products & Waste Products – Components**
  - ➢ **Product Cards**

- **Description Text**
- **Search / Filter UI**

---

11. Waste Products Page

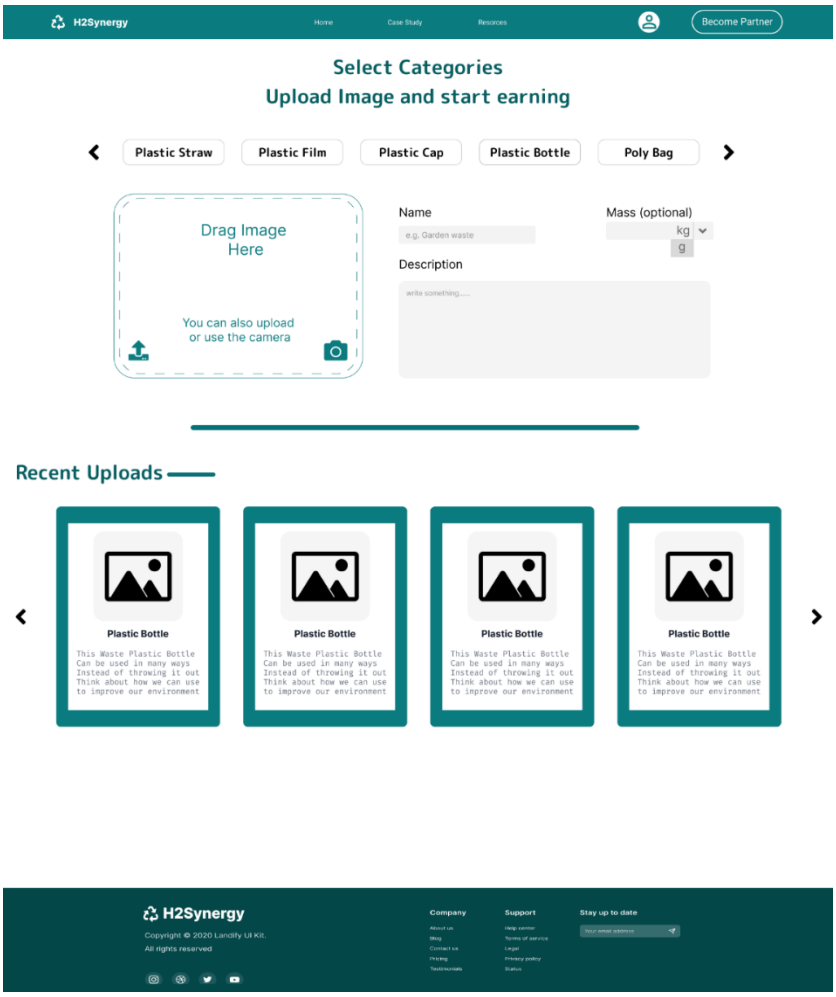
Displays a categorized list of accepted organic waste types. Users can learn about what kinds of waste yield the best hydrogen output and align their contributions accordingly.



12. Upload Waste

This interface allows users to upload images of their organic waste directly to the platform. Users can preview selected images, view tips for correct image formatting, and track the upload progress. Once submitted, the data is passed to the AI system for analysis.

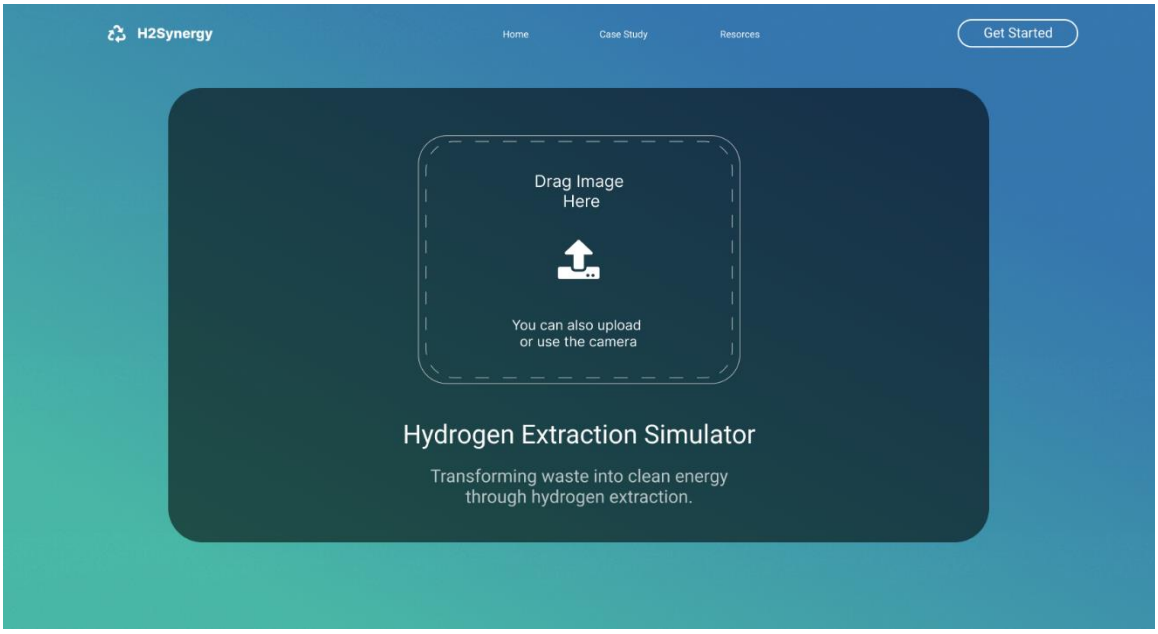
- **Upload Waste – Components**
  - ➢ **Image Upload Field**
  - ➢ **Preview Area**
  - ➢ **Upload Button**
  - ➢ **Progress Tracker**



13. AI Model Screen

This advanced screen is connected to the AI backend. It visually displays how uploaded waste is analyzed, including segmentation, classification, and volume prediction. It enhances user trust and transparency in the AI process.

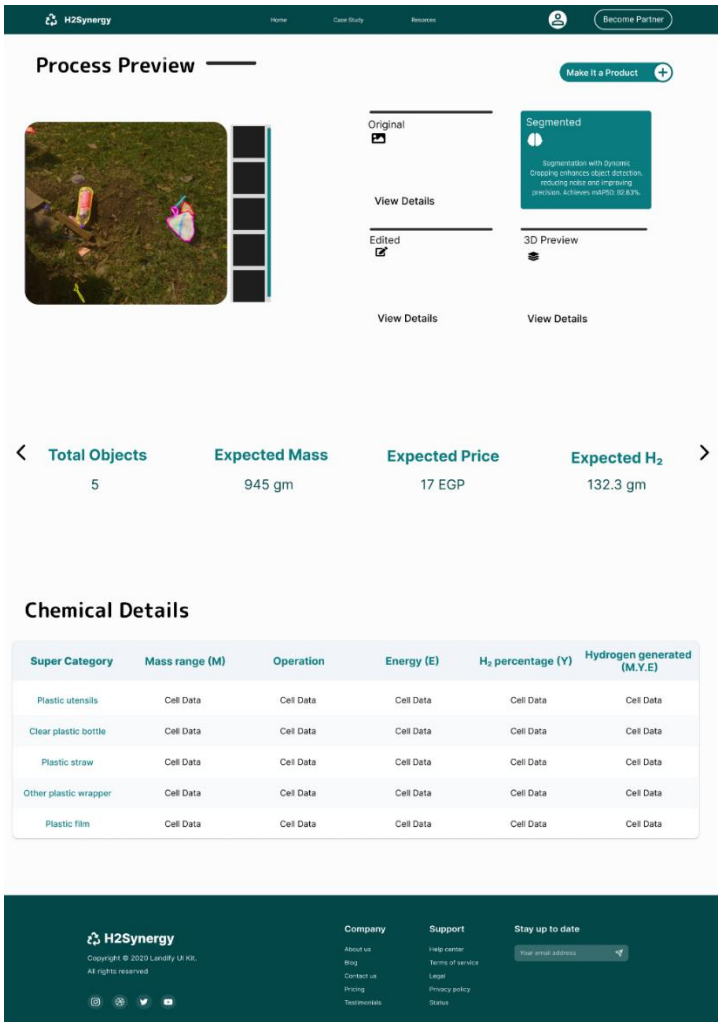
- **AI Model Screen – Components**
  - ➢ **Segmented Output Image**
  - ➢ **Heatmap / Analysis Overlay**
  - ➢ **Interpretation Text / Result Summary**



14. Results Page

After AI analysis, users are directed to the Results Page, where they can view waste classification results, predicted hydrogen output, and a summary of potential environmental impact. This interface emphasizes data clarity, using charts or indicators where applicable.

- **Results Page – Components**
  - ➢ **Hydrogen Output Card**
  - ➢ **Waste Classification Panel**
  - ➢ **Download Report Button**



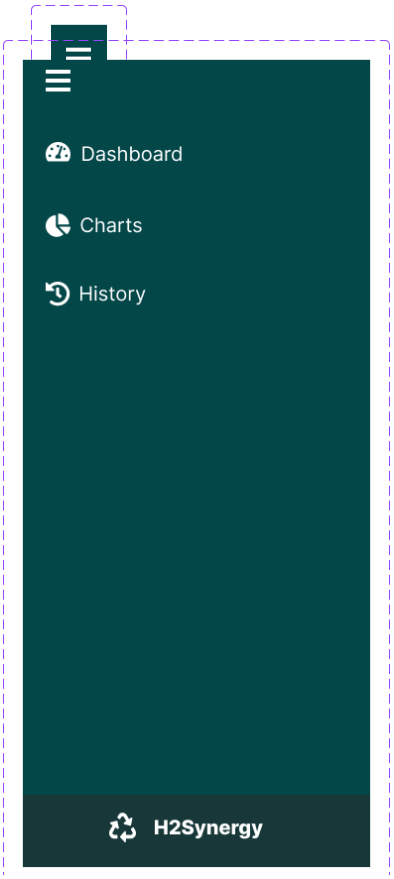
15. Navbar (Top Navigation)

The top navigation bar serves as a persistent interface element across the website, designed to enhance user experience through consistent and intuitive access to the platform's key sections. Positioned at the top of every page, it provides direct navigation to areas such as Home, Dashboard, Upload, Results, and Profile.

This component plays a critical role in usability by reducing navigation effort and ensuring that users, whether individuals or companies can easily switch between functional views without disrupting their workflow. Its visibility across all pages also supports platform familiarity and orientation, especially for new users.

- **Navbar – Key Components:**
- **Logo:** Positioned on the left side; it serves as a visual brand anchor and redirects to the Home Page when clicked.
- **Navigation Links:** Clearly labeled links to core pages like Home, Dashboard, and Profile. These are designed using responsive elements to adapt to various screen sizes.
- **User Dropdown Menu:** Appears when a user is logged in. It provides options such as View Profile, Settings, and Logout, supporting secure and personalized interactions.

16. Side Navigation (Collapsible Menu)



The side navigation panel is a collapsible vertical menu integrated into dashboard-related views of the platform, particularly after authentication. It allows users to manage their interactions with the system in an organized and task-specific manner. The design supports expand/collapse functionality, improving adaptability across devices and screen sizes.

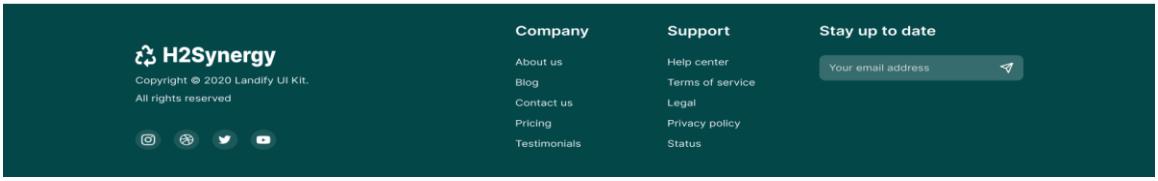
This sidebar enhances workflow by segmenting system features into clearly defined sections and offering immediate access through intuitive icons and labels. It is especially useful for users who frequently navigate between tasks such as uploading waste, viewing analysis results, or managing account details.

<u>Side Navigation – Key Components:</u>

- Collapsible Section Menu: A vertical menu that can expand to show full labels or collapse into icons only, optimizing screen space.

- Icons with Labels: Each menu item includes an icon and text label, enhancing accessibility and reducing cognitive load. Items typically include links to Upload, Results, History, and Settings.

17. Footer (Website Bottom Section)



The Footer is a persistent interface element located at the bottom of every page. It serves as a secondary navigation and information hub, reinforcing the credibility, accessibility, and completeness of the platform. While less prominent than the main navigation components, the footer provides essential links and legal information that users may refer to at any point in their interaction.

Its design is minimal and functional, ensuring that it does not distract from primary content while remaining easily accessible. The layout adapts responsively to various screen sizes, maintaining readability and utility across desktop and mobile views.

- **Footer – Key Components:**
  - **Quick Links**: A set of navigational shortcuts to core sections such as About Us, Contact, FAQ, Terms of Service, and Privacy Policy. These links support user trust and transparency.
  - **Social Media Icons**: Clickable icons linking to the platform's official pages on platforms like Facebook, Twitter, LinkedIn, or Instagram. These enhance engagement and brand presence.
  - **Copyright Notice**: A legal statement indicating ownership of the platform and its content (e.g., © 2025 Green Hydrogen Platform. All rights reserved).

The footer also contributes to search engine optimization (SEO) and site structure clarity, making it an important component in both the user interface and the technical architecture of the website.

4. Backend Architecture (Node.js)

The backend is developed using Node.js and Express.js to handle HTTP requests, authenticate users, process uploads, and interact with the AI engine.

Features:
- RESTful APIs for frontend-backend communication.
- MongoDB/PostgreSQL for storing user data, waste uploads, and results.
- JWT-based authentication for secure access.
- Error handling and data validation.
- Integration with AI model endpoints.

5. Deployment

The platform is deployed using modern DevOps practices:
- Frontend hosted on Vercel/Netlify.
- Backend hosted on Render/Heroku/Cloud server.
- Database managed on MongoDB Atlas or a cloud SQL provider.
- CI/CD pipelines ensure automatic deployment from GitHub.
- Environment variables are securely stored.

6. Integration Flow

1. User uploads organic waste image via frontend.
2. Image is sent to backend where it is processed and passed to AI module.
3. AI returns classification, segmentation, and predicted hydrogen output.
4. Backend stores results and sends them to frontend.
5. Frontend displays output in real time on the Results page.

7. Security & Scalability

- HTTPS enforced across the platform.
- JWT tokens for session management.
- Scalable React components and RESTful APIs.

- Backend supports horizontal scaling.
- Modular structure supports future AI model updates.

8. Project Contribution

The website serves as the main interface between the users and the green hydrogen system. It facilitates the collection of organic waste data, manages user interaction, and delivers AI-powered hydrogen predictions. By automating user engagement and processing, it plays a crucial role in ensuring sustainability and efficiency in the website.

Chapter 5

Results

5.1 Preprocessing and Augmentation for Segmentation Model:

Preprocessing	Auto-Orient: Applied
	Resize: Stretch to 608x608
	Filter Null: Require all images to contain annotations.
Augmentations	Outputs per training example: 3
	Flip: Horizontal, Vertical
	90° Rotate: Clockwise, Counter-Clockwise, Upside Down
	Crop: 0% Minimum Zoom, 15% Maximum Zoom
	Rotation: Between -8° and +8°
	Shear: ±3° Horizontal, ±3° Vertical
	Brightness: Between -20% and +20%
	Exposure: Between -10% and +10%

5.2 Segmentation Model Results:

This section presents the experimental results obtained from the integrated waste analysis pipeline, which encompasses semantic segmentation, object classification, volume estimation, weight computation, hydrogen yield estimation, and economic valuation. The system was evaluated on a representative sample of five waste objects, and the performance of each module is discussed in detail below.

The following files and components were used in this practical experiment to evaluate the proposed waste analysis pipeline:
- h_extraction_pipeline.py: contains the core pipeline logic for object detection, classification, volume, and weight estimation.
- results.json: includes processed output from the pipeline, including hydrogen and economic computations.
- material_properties.json: provides physical constants such as density, hydrogen ratio, and fill ratio for each material type.
- final_results.ipynb: notebook calls pipeline and contains all computation steps.

Figure (10) Test image

Semantic Segmentation Performance

The segmentation model was quantitatively evaluated using standard performance metrics, namely mean Average Precision at an Intersection over Union threshold of 0.5 (mAP@0.5), Precision, and Recall. The obtained results are as follows:

Metric	Value
mAP@0.5	64.1%
Precision	56.2%
Recall	63.9%

These values indicate that the segmentation model demonstrates reliable object localization capabilities. Qualitative assessment through the label visualization mask further supports the model's segmentation reliability, with distinct contours accurately delineating object boundaries.

Figure (11) Segmentation Output


```
[{'model_predictions': {'inference_id': '002d2b8c-0baf-4faf-acb7-016b31f04734',
   'predictions': {'image': {'width': 1000, 'height': 668},
    'predictions': [{'width': 166.0,
      'height': 55.0,
      'x': 382.0,
      'y': 586.5,
      'confidence': 0.9227487444877625,
      'class_id': 0,
      'points': [{'x': 302.0, 'y': 559.0},
       {'x': 300.0, 'y': 561.0},
       {'x': 300.0, 'y': 562.0},
       {'x': 299.0, 'y': 563.0},
       {'x': 298.0, 'y': 563.0},
       {'x': 298.0, 'y': 605.0},
       {'x': 300.0, 'y': 605.0},
       {'x': 301.0, 'y': 606.0},
```

5.3 Object Classification Outcomes

Following segmentation, each object was classified using either the primary classification model or a specialized classifier for food waste, depending on the identified category. In this test scenario, four objects were classified as "clear plastic bottles," and one as a "plastic bottle cap." No food waste was detected; thus the secondary classification stage was not invoked.

5.4 Volume Estimation

Volume estimation was conducted using a structured reasoning framework powered by Mask R-CNN Fine-Tuned model. In cases where volumetric inference was not feasible, empirically derived class-based average volumes were used.

Estimated Volumes of Detected Objects:

Object ID	Class	Estimated Volume (ml)
20250608201433187866	Clear Plastic Bottle	500.0
20250608201433143006	Clear Plastic Bottle	750.0
20250608201433062891	Clear Plastic Bottle	770.0
20250608201433177523	Plastic Bottle Cap	0.0
20250608201433148983	Clear Plastic Bottle	770.0

5.5 Weight Estimation

Weight was computed using the formula: Weight (g) = Volume (ml) × Density (g/ml) × Fill Ratio

Weight Estimations:

Object ID	Class	Estimated Weight (g)
20250608201433187866	Clear Plastic Bottle	17.79
20250608201433143006	Clear Plastic Bottle	26.6
20250608201433062891	Clear Plastic Bottle	27.39
20250608201433177523	Plastic Bottle Cap	0.00
20250608201433148983	Clear Plastic Bottle	27.39

5.6 Hydrogen Yield Estimation

Hydrogen yield was calculated using: Hydrogen Yield (g) = Weight (g) × Hydrogen Ratio

Hydrogen Yield Estimates:

Object ID	Class	Hydrogen Yield (g)
20250608201433187866	Clear Plastic Bottle	2.49
20250608201433143006	Clear Plastic Bottle	3.74
20250608201433062891	Clear Plastic Bottle	3.84
20250608201433177523	Plastic Bottle Cap	0.00
20250608201433148983	Clear Plastic Bottle	3.84

5.7 Economic Valuation

Assuming a market price of $0.0045 USD per 1 gram of hydrogen, the total value of 30.43 grams of hydrogen is calculated as:

Total Value (USD) = (13.91× 0.0045) = $0.07

Figure (12) Sample Output


```
{'20250614135536093318': {'volume_ml': 500.0,
  'class': 'clear plastic bottle',
  'segmentation_confidence': 0.92,
  'volume_est_confidence': '0.75',
  'density': 1.39,
  'hydrogen_ratio': 0.14,
  'fill_ratio': 0.032,
  'expected_mass': 17.792,
  'total_extracted_hydrogen': 2.49,
  'total_extracted_hydrogen_in_kg': 0.002},
 '20250614135536116722': {'volume_ml': 750.0,
  'class': 'clear plastic bottle',
  'segmentation_confidence': 0.86,
  'volume_est_confidence': '0.75',
  'density': 1.39,
  'hydrogen_ratio': 0.14,
  'fill_ratio': 0.032,
  'expected_mass': 26.688000000000002,
  'total_extracted_hydrogen': 3.74,
  'total_extracted_hydrogen_in_kg': 0.004},
 '20250614135536149262': {'volume_ml': 770.0,
  'class': 'clear plastic bottle',
  'segmentation_confidence': 0.77,
  'volume_est_confidence': 0.75,
  'density': 1.39,
  'hydrogen_ratio': 0.14,
  'fill_ratio': 0.032,
  'expected_mass': 27.399680000000004,
  'total_extracted_hydrogen': 3.84,
  'total_extracted_hydrogen_in_kg': 0.004},
 '20250614135536158940': {'volume_ml': 0.0,
  'class': 'plastic bottle cap',
  'segmentation_confidence': 0.66,
  'volume_est_confidence': '1.0',
  'density': 0.95,
  'hydrogen_ratio': 0.14,
  'fill_ratio': 0.3,
  'expected_mass': 0.0,
  'total_extracted_hydrogen': 0.0,
  'total_extracted_hydrogen_in_kg': 0.0},
```

```
'20250614135536164492': {'volume_ml': 770.0,
 'class': 'clear plastic bottle',
 'segmentation_confidence': 0.56,
 'volume_est_confidence': '0.6',
 'density': 1.39,
 'hydrogen_ratio': 0.14,
 'fill_ratio': 0.032,
 'expected_mass': 27.399680000000004,
 'total_extracted_hydrogen': 3.84,
 'total_extracted_hydrogen_in_kg': 0.004}}
```


Figure (13) Summary of results


```
{'number_of_objects': 5,
 'total_hydrogen': 13.91,
 'total_expected_mass': 99.27936000000001,
 'total_objects': 5,
 'total_energy': 0.49000000000000005,
 'total_cost_in_usd': 0.07}
```

5.9 Evaluation

We present the practical evaluation of our system using real images of organic waste collected from actual environments. This evaluation aims to test the system's performance in estimating the weight and hydrogen content of different waste types under realistic conditions.

Experimental Setup

To begin the evaluation, we used a real-world test image containing multiple organic waste items placed naturally in a single frame. This image served as the input for the system.

Figure (14) Test image

The following steps outline how each object within this image was processed through the pipeline, starting from segmentation and classification to weight and hydrogen estimation.

Segmentation Model Results:

The system first performed segmentation to identify and isolate individual waste items from the image. Below is the output showing the segmented regions and their corresponding classes.

Figure (15) Segmentation model output as images

Figure (16) Segmentation model output in the Python notebook


```
[{'model_predictions': {'inference_id': '8e28aa40-6414-43ae-bbbe-469984323992',
   'predictions': {'image': {'width': 1083, 'height': 1059},
    'predictions': [{'width': 83.0,
      'height': 47.0,
      'x': 540.5,
      'y': 211.5,
      'confidence': 0.945336103439331,
      'class_id': 12,
      'points': [{'x': 515.0, 'y': 188.0},
       {'x': 514.0, 'y': 189.0},
       {'x': 509.0, 'y': 189.0},
       {'x': 503.0, 'y': 195.0},
       {'x': 501.0, 'y': 195.0},
       {'x': 500.0, 'y': 196.0},
       {'x': 500.0, 'y': 230.0},
       {'x': 506.0, 'y': 224.0},
       {'x': 509.0, 'y': 224.0},
       {'x': 510.0, 'y': 223.0},
       {'x': 539.0, 'y': 223.0},
       {'x': 540.0, 'y': 224.0},
       {'x': 543.0, 'y': 224.0},
       {'x': 544.0, 'y': 225.0},
       {'x': 545.0, 'y': 225.0},
       {'x': 547.0, 'y': 227.0},
       {'x': 552.0, 'y': 227.0},
       {'x': 553.0, 'y': 228.0},
       {'x': 568.0, 'y': 228.0},
       {'x': 570.0, 'y': 230.0},
       {'x': 575.0, 'y': 230.0},
```

Classification Model Results:

After segmentation, the system classified the object using the trained model.

Figure (17) Classification results

Volume and Weight Estimation Results:

Following the classification stage, the system proceeded to **estimate the volume** of each identified object using the segmented regions. These volume values served as the foundation for subsequent calculations.

Using the estimated volume, along with the **known density** and **fill ratio** for each material type, the **expected weight** of each object was calculated. This allowed the system to approximate the physical mass of the waste items in a realistic manner.

Building on that, we applied the **hydrogen ratio** associated with each material to determine the **amount of extractable hydrogen** per object. Finally, the system estimated the **economic value** of the recovered hydrogen based on its quantity.

The results below represent the output of these stages in the pipeline, applied to each object in the real-world image.

Figure (18) Results of Volume and Weight Estimation


```
{'20250614133909283813': {'volume_ml': 9.0,
  'class': 'plastic bottle cap',
  'segmentation_confidence': 0.95,
  'volume_est_confidence': '0.85',
  'density': 0.95,
  'hydrogen_ratio': 0.14,
  'fill_ratio': 0.3,
  'expected_mass': 2.0519999999999996,
  'total_extracted_hydrogen': 0.29,
  'total_extracted_hydrogen_in_kg': 0.0},
 '20250614133909350294': {'volume_ml': 14.0,
  'class': 'plastic bottle cap',
  'segmentation_confidence': 0.91,
  'volume_est_confidence': '0.85',
  'density': 0.95,
  'hydrogen_ratio': 0.14,
  'fill_ratio': 0.3,
  'expected_mass': 3.1919999999999997,
  'total_extracted_hydrogen': 0.45,
  'total_extracted_hydrogen_in_kg': 0.0},
 '20250614133909362902': {'volume_ml': 500.0,
  'class': 'clear plastic bottle',
  'segmentation_confidence': 0.88,
  'volume_est_confidence': '0.85',
  'density': 1.39,
  'hydrogen_ratio': 0.14,
  'fill_ratio': 0.032,
  'expected_mass': 17.792,
  'total_extracted_hydrogen': 2.49,
  'total_extracted_hydrogen_in_kg': 0.002},
 '20250614133909515997': {'volume_ml': 10.0,
  'class': 'plastic bottle cap',
  'segmentation_confidence': 0.86,
  'volume_est_confidence': '0.85',
  'density': 0.95,
  'hydrogen_ratio': 0.14,
  'fill_ratio': 0.3,
  'expected_mass': 2.2800000000000002,
  'total_extracted_hydrogen': 0.32,
  'total_extracted_hydrogen_in_kg': 0.0},
```

```
 '20250614133909522162': {'volume_ml': 1000.0,
  'class': 'clear plastic bottle',
  'segmentation_confidence': 0.63,
  'volume_est_confidence': '0.85',
  'density': 1.39,
  'hydrogen_ratio': 0.14,
  'fill_ratio': 0.032,
  'expected_mass': 35.584,
  'total_extracted_hydrogen': 4.98,
  'total_extracted_hydrogen_in_kg': 0.005},
 '20250614133909767034': {'volume_ml': 1000.0,
  'class': 'clear plastic bottle',
  'segmentation_confidence': 0.51,
  'volume_est_confidence': '0.8',
  'density': 1.39,
  'hydrogen_ratio': 0.14,
  'fill_ratio': 0.032,
  'expected_mass': 35.584,
  'total_extracted_hydrogen': 4.98,
  'total_extracted_hydrogen_in_kg': 0.005}}
```

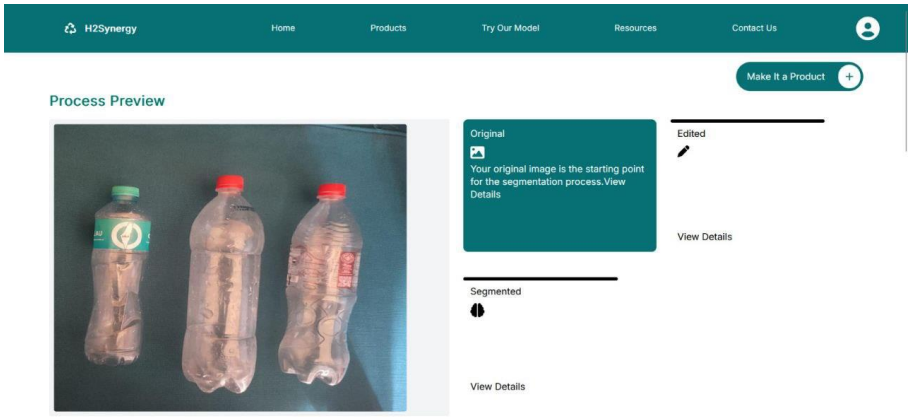

Figure (19) Total results for all objects


```
{'number_of_objects': 6,
 'total_hydrogen': 13.510000000000002,
 'total_expected_mass': 96.48400000000001,
 'total_objects': 6,
 'total_energy': 0.43,
 'total_cost_in_usd': 0.05}
```

Website Results:

After the Image is uploaded, the Results and the Process reviews will appear in this page, clarify general information and Chemical Details with saving option

Figure (20) UI/UX results Page.



Id	class	expected_mass	total_extracted_hydrogen	volume_ml
20250613172707136940	plastic bottle cap	2.05	0.29	9
20250613172707196366	plastic bottle cap	3.19	0.45	14
20250613172707242212	clear plastic bottle	17.792	2.49	500
20250613172707291293	plastic bottle cap	2.28	0.32	10
20250613172707295264	clear plastic bottle	35.584	4.98	1000
20250613172707415622	clear plastic bottle	35.584	4.98	1000

6. Conclusion:

This graduation project proposes a comprehensive and technically sound framework for estimating the **weight**, **hydrogen content**, and **economic value** of organic waste objects based solely on image input. The proposed system integrates deep learning models with domain-specific physical and chemical knowledge to deliver meaningful and actionable insights from simple visual data.

The pipeline begins with the application of a **volume estimation model**, which was trained to infer the approximate volume of an object from a 2D image input. This model leverages convolutional neural networks (CNNs) to analyze object geometry, contours, and scale relationships within the scene, enabling it to provide accurate volumetric estimates even when working with single-view images. This step forms the foundation of all subsequent estimations in the system.

Once the volume of each object is estimated, it is passed into a structured calculation module that incorporates a set of predefined physical constants for each material type. These include:

- **Density (g/cm^3)**: the average mass per unit volume of the object's material.
- **Fill ratio**: the estimated proportion of the object's interior that is actually filled with material (to account for hollow or partially filled items).
- **Hydrogen ratio**: the average percentage of the object's weight that corresponds to hydrogen content, derived from chemical composition studies.

Using these constants and the estimated volume, the system applies rule-based equations to compute the following values:

- **Expected weight (mass)** of the object in grams.
- **Extractable hydrogen content**, both in grams and in kilograms.

- **Estimated monetary value** based on the amount of hydrogen and its assumed energy or commercial value.

These estimations are computed per object and then aggregated across all items in the image to provide a total expected output. The system was tested using a real-world image containing multiple types of organic waste. The same pipeline was applied across all items consistently, and the results were compiled into structured tables that reflect the system's performance.

The evaluation showed that the combined use of a volume estimation neural network and deterministic material-based calculations leads to reliable and interpretable outputs. The framework is entirely modular, allowing each stage to be improved independently. For example, the volume estimation model can be retrained or fine-tuned with a larger and more diverse dataset to support additional waste categories. Similarly, the database of material constants can be expanded or dynamically adjusted based on regional data or updated hydrogen market values.

In terms of sustainability impact, this project demonstrates the potential of lightweight, image-based modeling to support decision-making in waste management and green hydrogen production. By providing a fast, scalable, and non-invasive method to evaluate the value of waste, the system offers a promising tool for industrial, municipal, and research applications. It helps bridge the gap between unstructured waste and structured energy potential, contributing directly to the goals of the circular economy and clean energy transition.

7. Data Availability:

Dataset:

https://www.kaggle.com/datasets/bjoernjostein/food-classification

Segmentation Model:

https://universe.roboflow.com/taco-9911u/taco-trash-annotations-in-context-bumvw/model/15

Classification Model:

https://paperswithcode.com/method/mobilenetv2

Volume Estimation Model:

https://github.com/AlexGraikos/food_volume_estimation

Github For Our Project:
https://github.com/YousefMohammad/clean_hydro_extraction

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